OBSTRUCTION RESIDUES FOR THE $3N-1$ MAP: FULL FACTOR COMPLEXITY, POSITIVE DENSITY, AND A CONSTRUCTIVE LIFT

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Abstract. We study a family $\mathcal{O}_L \subseteq \{1,\dots,2^L\}$ of residue classes modulo 2^L arising naturally in the dynamics of the $3n-1$ map $T_-(n) := (3n-1)/2^{v_2(3n-1)}$. The family is defined by an algebraic criterion on a two-track parallel reduction. We prove three structural results: a constructive lift between modular levels, the existence of a positive asymptotic density c_W with rigorous lower bound $c_W \geq 7556/65536 \approx 0.115$, and full factor complexity $p_W(\ell) = 2^\ell$ of the associated language of binary representations. The central technical tool is a parity lemma in the residue system, which forces the terminal data of the parallel reduction into a shift-s normal form with integer-valued shift index. A finer classification by shift index yields a strict lift balance, an exact series representation of c_W, and reduces the open question of c_W's exact value to a single quantitative bound on a sub-family of atomic obstructions. Results transfer to the classical $3n+1$ map via a residue-level involution $r \mapsto 2^L - r$ that maps the T_- parallel reduction to its T_+ analogue step-by-step.

1. Introduction

1.1. **The Collatz map and its $3n-1$ cousin.** Let $\mathbb{N}_{\text{odd}}$ denote the set of positive odd integers, and let $v_2(m)$ denote the 2-adic valuation of an integer $m \neq 0$. The *Collatz map* T_+ takes an odd integer n to the odd part of $3n+1$:

$$T_+(n) := \frac{3n+1}{2^{v_2(3n+1)}}.$$

The Collatz conjecture, attributed to Lothar Collatz and widely studied since the 1930s, asserts that for every $n \in \mathbb{N}_{\text{odd}}$ the iteration $n, T_+(n), T_+^2(n), \dots$ eventually reaches the fixed point 1. Despite considerable computational evidence and a large body of partial results — early density-side analyses by Terras [Ter76] and Everett [Eve77], refined density estimates by Crandall [Cra78] and Korec [Kor94], and most recently the breakthrough of Tao [Tao22], showing that almost all trajectories attain almost bounded values — the conjecture itself remains open. We refer the reader to the

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annotated bibliographies of Lagarias [Lag10b, Lag10a] for a comprehensive overview of the literature.

The Collatz map has a natural cousin, the $3n-1$ *map* T_- acting on the same domain:

$$T_-(n) := \frac{3n-1}{2^{v_2(3n-1)}}.$$

The two maps are structurally parallel: both perform the operation "multiply by three, add a small constant, strip the factors of two from the result". They differ only in the sign of the additive constant. The trajectories of T_- are, however, *not* believed to converge to a single attractor: three cycles are classically known — the trivial fixed cycle through 1, the $\{5,7\}$-cycle, and the 7-cycle $17 \to 25 \to 37 \to 55 \to 41 \to 61 \to 91 \to 17$; see Lagarias [Lag10a] for historical attribution. The question of whether further cycles exist remains as open as the Collatz conjecture itself. The two maps are connected through a classical identity going back to Lagarias [Lag85]: every T_--trajectory starting from m corresponds, via the substitution $r = 2m+1$, to a single T_+-step followed by a T_+-trajectory. This *Lagarias intertwining* illustrates that any structural feature of T_- has a T_+ counterpart; the explicit transfer we use in Section 7, however, is different — a residue-level involution $r \mapsto 2^L - r$ that conjugates the T_- parallel reduction into the T_+ parallel reduction step-by-step (Lemma 7.2).

The present paper is concerned with a family of residue classes modulo 2^L that arises naturally in the dynamics of T_-. We show that this family — which we shall call the *obstruction residues* for reasons made precise in Section 2 — has three structural properties:

(1) a constructive lift between modular levels L and $L+1$,
(2) positive asymptotic density in $\{1,\dots,2^L\}$ as $L \to \infty$,
(3) maximal combinatorial complexity of its associated language of bit strings.

The third property is in some tension with the first two: a constructive lift and positive density would, in many natural settings, force a certain rigidity of the resulting word language (one might expect a sofic shift or a primitive substitution shift). We prove that no such substitution-type characterization is possible, identifying the obstruction residues as a structurally richer family than the naive analogies would suggest.

1.2. **Parallel reductions and the X-invariant.** The construction is built on a simple observation. For an odd integer r with $v := v_2(r-1) \geq 1$, write $m := (r-1)/2^v$. We run T_- in parallel on r and on m modulo 2^L for a fixed level L, keeping track at each step both of the current values and of their *affine relationship*: a pair $(c_j, d_j) \in \mathbb{Q}^2$ such that $T_-^j(m) \equiv c_j \cdot T_-^j(r) + d_j$ modulo an appropriately reduced power of two. The pair is updated by an explicit rule depending on the 2-adic valuations of the two trajectories. We denote the terminal pair $(c_{\text{end}}, d_{\text{end}})$ and define the scalar invariant

$$X_{\text{end}}(r, L) := 3d_{\text{end}}(r, L) + c_{\text{end}}(r, L).$$

For an odd integer r with $v_2(r-1) \geq 1$ and a level $L > v_2(r-1)$, we call r an *obstruction residue at level* L when $X_{\mathrm{end}}(r, L) = 1$, and write $\mathcal{O}_L \subseteq \{1, \ldots, 2^L\}$ for the set of all such residues. Definition 2.2 in Section 2 restates this with the parallel reduction made explicit; the typical case is $v = 1$, i.e. $r \equiv 3 \pmod 4$.

The family $(\mathcal{O}_L)_{L \geq 5}$ is small at first — $\mathcal{O}_5 = \{27\}$, $\mathcal{O}_6 = \{19, 27, 59\}$ — but grows rapidly: direct enumeration gives

$$|\mathcal{O}_5| = 1, \quad |\mathcal{O}_6| = 3, \quad |\mathcal{O}_7| = 8, \quad |\mathcal{O}_8| = 19, \quad \ldots, \quad |\mathcal{O}_{16}| = 7556,$$

and the density ratio $|\mathcal{O}_L|/2^L$ rises monotonically. Full counts at all levels $L = 5, \ldots, 16$ together with the shift-zero / shift-nonzero and atomic-shift-nonzero decompositions are collected in Table 3 (Appendix A). Three structural questions arise:

(Q1) Does $r_0 \in \mathcal{O}_{L_0}$ guarantee that both r_0 and $r_0 + 2^{L_0}$ lie in $\mathcal{O}_{L_0+1}$?
(Q2) Does $|\mathcal{O}_L|/2^L$ converge to a positive limit, and can the limit be bounded from below?
(Q3) What is the set of binary words appearing as factors of the bit representations of residues in $\bigcup_L \mathcal{O}_L$?

The three questions are not independent: a positive answer to (Q1) gives monotonicity, hence most of (Q2). A constructive form of (Q1), exhibiting arbitrary bit patterns under lift, gives (Q3). All three are answered in the affirmative by the main results; in particular the factor language $\mathcal{F}(\mathcal{O})$ defined below turns out to be all of $\{0,1\}^*$.

1.3. **Main results.**

Theorem 1.1 (Constructive lift)**.** *Let $r_0 \in \mathcal{O}_{L_0}$. Then both $r_+ := r_0$ and $r_- := r_0 + 2^{L_0}$ are obstruction residues at level $L_0 + 1$.*

The proof, given in Section 3, proceeds by case analysis on a finite invariant of the parallel reduction. (No explicit lower bound on L_0 is needed: for $L_0 \leq 4$ the family $\mathcal{O}_{L_0}$ is empty (Remark 2.3) and the statement is vacuous; the smallest non-vacuous level is $L_0 = 5$ with $\mathcal{O}_5 = \{27\}$. The proof in Section 3 only uses $L_0 > v$, which is automatic from $r_0 \in \mathcal{O}_{L_0}$ by Definition 2.2.) The lift is explicit: for each r_0, the terminal data of the reduction at r_+ or r_- is determined by a closed-form algebraic identity.

Theorem 1.2 (Positive density)**.** *The density ratio $|\mathcal{O}_L|/2^L$ is non-decreasing in L, and the limit*

$$c_W := \lim_{L \to \infty} \frac{|\mathcal{O}_L|}{2^L}$$

exists in $(0, 1]$. For every $L_0 \geq 5$, the rigorous lower bound $c_W \geq |\mathcal{O}_{L_0}|/2^{L_0}$ holds; at $L_0 = 16$ this gives $c_W \geq 7556/65536 \approx 0.115$, sharpening the conservative bound $c_W \geq 3/64 \approx 0.047$ obtained from $L_0 = 6$.

The bound at $L_0 = 16$, using only the algorithmic X_{end}-criterion of Section 2, gives the explicit value $7556/65536 \approx 0.115$ stated above. Raising

L_0 further admits substantial additional improvement — direct enumeration at $L_0 = 32$ already yields $c_W \geq \rho_{32} \approx 0.147$ (Table 1) — but the exact value of c_W remains an open problem. We give in Section 6 a refined decomposition that exhibits c_W as a rigorously convergent series and isolates the remaining open question to a single quantitative bound on a sub-family of obstructions.

For $r \in \mathcal{O}_L$, write $w_r \in \{0,1\}^L$ for the binary representation of r as an L-bit word, least significant bit first. The *factor language* of the obstruction family is

$$\mathcal{F}(\mathcal{O}) := \{\, u \in \{0,1\}^* : u \text{ is a factor of } w_r \text{ for some } r \in \mathcal{O}_L,\ L \geq 5 \,\},$$

and the *factor complexity function* is $p_W(\ell) := |\mathcal{F}(\mathcal{O}) \cap \{0,1\}^\ell|$.

Theorem 1.3 (Full factor complexity)**.** *For every $\ell \geq 0$,*

$$p_W(\ell) = 2^\ell.$$

The proof, in Section 5, is constructive: for each binary word u, we exhibit an explicit residue $r(u)$ in some $\mathcal{O}_L$ having u as a factor. The construction is a direct iteration of the lift theorem.

A consequence of Theorem 1.3 is a sharp structural rigidity (Corollary 5.2): the obstruction family cannot be described as a proper subshift of finite type, a proper sofic shift, or a primitive substitution shift in the sense of Pansiot [Pan84] or Devyatov [Dev15].

1.4. **Strategy.** All three theorems rest on a single algebraic structure: the *shift-s normal form* of the terminal data. We prove in Section 2 (Lemma 2.5) that whenever the parallel reduction terminates with $X_{\text{end}} = 1$, the terminal pair has the form

$$(c_{\text{end}}, d_{\text{end}}) = \left(4^s, \frac{1-4^s}{3}\right) \qquad \text{for some } s \in \mathbb{Z}.$$

The integer s, which we call the *shift index*, measures the difference between the 2-adic valuations of the two parallel trajectories. Theorem 1.1 is proved by case analysis on how the shift index evolves under lift; Theorem 1.3 is the iterative consequence; Theorem 1.2 follows by monotonicity from the lift. A refined classification by shift index (Section 6) then re-expresses c_W as a convergent series and isolates the remaining open question.

The shift-s normal form is, to the best of our knowledge, new; in particular, it differs from the Markov-operator setup of Wirsching [Wir98], which works with an invariant probability-density function rather than a terminal algebraic datum. The remaining analysis proceeds by elementary case analysis and combinatorial bookkeeping; the arguments do not depend on specialized Collatz machinery beyond the Lagarias intertwining.

1.5. **Outline.** Section 2 introduces the parallel reduction, defines the X-invariant rigorously, and proves the parity lemma. Section 3 contains the proof of Theorem 1.1, split into the two stop-type cases. Section 4 derives Theorem 1.2 as a direct corollary. Section 5 proves Theorem 1.3 and derives the structural rigidity corollary. Section 6 develops the classification by shift index, proves the strict lift balance and series representation, and isolates the remaining open problem. Section 7 transfers the results to the $3n+1$ side via the residue bijection. Section 8 discusses open questions. Appendix A summarizes computational verifications.

2. The parallel reduction and the X-invariant

2.1. **The basic step.** Recall that for an odd integer n, the map T_- strips the 2-adic part from $3n-1$:

$$T_-(n) = \frac{3n-1}{2^{v_2(3n-1)}}.$$

We need a version of this step that keeps track of the modulus. Fix a level $L \geq 1$, and consider a pair (α, β) with α a positive odd integer and β a power of two, interpreted as a residue α modulo β. A single T_--step on such a pair produces

$$(\alpha', \beta') = \left(\frac{3\alpha - 1}{2^{v_\alpha}}, \frac{3\beta}{2^{v_\alpha}}\right), \qquad v_\alpha := v_2(3\alpha - 1),$$

provided $v_\alpha < v_2(\beta)$. If $v_\alpha \geq v_2(\beta)$, the modulus has been exhausted and the step is undefined; we say the reduction has *terminated*.

Since 3 is odd, $v_2(3\beta) = v_2(\beta)$, so $v_2(\beta') = v_2(\beta) - v_\alpha$: the modulus strictly decreases at each step, and the reduction always terminates after finitely many steps. Iterates of β are in general of the form $3^j \cdot 2^k$ rather than literal powers of two; only the 2-adic valuation $v_2(\beta)$ is used downstream, so we track β as a positive integer of known 2-adic valuation.

2.2. **The two-track setup.** Fix r a positive odd integer with $r \neq 1$, set $v := v_2(r-1)$, and write $m := (r-1)/2^v$; then m is odd. The inequality $v \geq 1$ is automatic for odd $r > 1$ since $r - 1$ is then even. (We use v as a parameter throughout; the special case $v = 1$ — equivalently $r \equiv 3 \pmod 4$ — is the typical reader picture and is presented first wherever a concrete calculation helps.)

Fix a level $L > v$. We run two T_--reductions in parallel:

- the *K-track*, starting from $(a_K^{(0)}, b_K^{(0)}) := (r, 2^L)$,
- the *I-track*, starting from $(a_I^{(0)}, b_I^{(0)}) := (m, 2^{L-v})$.

The initial relation $m = (r-1)/2^v$ writes as $a_I^{(0)} = 2^{-v} a_K^{(0)} - 2^{-v}$. At each step $j \geq 0$, we update both tracks according to the basic step, denoting $v_K^{(j)} := v_2(3a_K^{(j)} - 1)$ and $v_I^{(j)} := v_2(3a_I^{(j)} - 1)$. Cumulative valuations along the two tracks are denoted $V_K^{(j)} := \sum_{i<j} v_K^{(i)}$ and $V_I^{(j)} := \sum_{i<j} v_I^{(i)}$, so that the residual 2-adic valuation of $b_K^{(j)}$ is $L - V_K^{(j)}$ and that of $b_I^{(j)}$ is $L - v - V_I^{(j)}$. The reduction proceeds as long as both tracks can step; it terminates as soon as either condition fails. We write $J = J(r, L)$ for the index of termination. In particular, once a track's cumulative valuation reaches $V_K^{(j)} \geq L - 1$ on

the K-track (or $V_I^{(j)} \geq L - v - 1$ on the I-track), no further step is possible there: the residual modulus valuation is then ≤ 1, while the relevant $a^{(j)}$ is odd, so $3a^{(j)} - 1$ is even, the step valuation is ≥ 1, and the stop condition is met. We invoke this *no-further-step fact* wherever a lift or a child reduction adds at most one further step.

2.3. The affine relation and the X-invariant.

Lemma 2.1. *For each j with $0 \leq j \leq J$, there exist $c_j, d_j \in \mathbb{Q}$ such that*

$$a_I^{(j)} = c_j\, a_K^{(j)} + d_j \qquad \text{in } \mathbb{Q},$$

with $c_0 = 2^{-v}$, $d_0 = -2^{-v}$, and the update rule

$$c_{j+1} = c_j \cdot 2^{v_K^{(j)} - v_I^{(j)}}, \qquad d_{j+1} = \frac{3d_j + c_j - 1}{2^{v_I^{(j)}}}.$$

Proof. The base case is immediate. Assume the affine relation holds at index j. Writing $A_K^{(j)} := 3a_K^{(j)} - 1$, $A_I^{(j)} := 3a_I^{(j)} - 1$, and $X_j := 3d_j + c_j$,

$$\begin{aligned} A_I^{(j)} &= 3(c_j a_K^{(j)} + d_j) - 1 \\ &= c_j(3a_K^{(j)} - 1) + (3d_j + c_j - 1) \\ &= c_j A_K^{(j)} + (X_j - 1). \end{aligned}$$

Dividing by $2^{v_I^{(j)}}$ and using $A_K^{(j)} = 2^{v_K^{(j)}} a_K^{(j+1)}$:

$$a_I^{(j+1)} = c_j \cdot 2^{v_K^{(j)} - v_I^{(j)}}\, a_K^{(j+1)} + \frac{X_j - 1}{2^{v_I^{(j)}}},$$

which is the desired affine relation at index $j + 1$. $\square$

We define the X*-invariant* of the parallel reduction at (r, L) as

$$X_{\mathrm{end}}(r, L) := X_J = 3d_J + c_J.$$

This is not invariant under the update step; the name reflects its role as the obstruction certificate (Definition 2.2).

2.4. Definition of obstruction residues.

Definition 2.2 (Obstruction residue)**.** Let r be a positive odd integer with $r \neq 1$, and write $v := v_2(r - 1) \geq 1$. We call r an *obstruction residue at level* $L > v$ if $X_{\mathrm{end}}(r, L) = 1$. We write

$$\mathcal{O}_L := \left\{ r \in \{1, \ldots, 2^L\} \;\middle|\; \begin{array}{l} r \text{ odd},\ v_2(r-1) \geq 1, \\ L > v_2(r-1),\ X_{\mathrm{end}}(r, L) = 1 \end{array} \right\}.$$

The definition is algorithmic: given r and L, one runs the parallel reduction, observes the terminal pair (c_J, d_J), and checks $3d_J + c_J = 1$. The computation terminates in at most L steps.

Remark 2.3. A direct enumeration confirms $\mathcal{O}_L = \emptyset$ for $1 \le L \le 4$; the smallest level at which an obstruction residue exists is $L = 5$, with $\mathcal{O}_5 = \{27\}$. At the next level, $\mathcal{O}_6 = \{19, 27, 59\}$, comprising the new atomic residue 19 (an obstruction that is not a lift of any lower-level obstruction, the notion made precise in Section 6 below) and the two lifts $\{27, 59\}$ of $27 \in \mathcal{O}_5$.

We call such r *obstruction residues* because $X_{\text{end}}(r, L) = 1$ is exactly the rigidity that forces the two parallel reductions to terminate in lockstep (Corollary 2.6). It also collapses their terminal affine relation to the shift-s normal form developed below; a generic residue exhibits no such coupling. The name also has a dynamical reading at the level of T_--trajectories rather than the parallel reduction; this reading motivates the definition but is developed elsewhere, and the present paper uses only the algebraic side established above.

2.5. **A worked instance.**

Example 2.4 (The reduction at $r = 27$, and a non-obstruction)**.** Take $r = 27$ at level $L = 5$, so $v = v_2(26) = 1$ and $m = 13$. Running the two tracks (with $v_K^{(j)} := v_2(3a_K^{(j)} - 1)$ and $v_I^{(j)} := v_2(3a_I^{(j)} - 1)$, and recording the affine data of Lemma 2.1):

j	$a_K^{(j)}$	$a_I^{(j)}$	$v_K^{(j)}$	$v_I^{(j)}$	c_j	d_j
0	27	13	4	1	$\frac{1}{2}$	$-\frac{1}{2}$
1	5	19	1	3	4	-1

Both tracks stop at $j = 1$, each step valuation reaching the residual valuation of its modulus — $L - V_K^{(j)}$ on the K-track, $(L - v) - V_I^{(j)}$ on the I-track (note the extra $-v$):

$$v_K^{(1)} = 1 \ge L - V_K^{(1)} = 5 - 4 = 1, \qquad v_I^{(1)} = 3 \ge (L - v) - V_I^{(1)} = 4 - 1 = 3.$$

The terminal pair is $(c_J, d_J) = (4, -1)$, so $X_{\text{end}}(27, 5) = 3 \cdot (-1) + 4 = 1$ and $27 \in \mathcal{O}_5$.

By contrast, $r = 7$ at $L = 5$ (here $v = 1$, $m = 3$) stops at $J = 1$ with terminal pair $(c_J, d_J) = (\frac{1}{4}, -\frac{1}{4})$, giving

$$X_{\text{end}}(7, 5) = 3 \cdot (-\tfrac{1}{4}) + \tfrac{1}{4} = -\tfrac{1}{2} \ne 1,$$

so $7 \notin \mathcal{O}_5$. The pair separates a necessary condition from a sufficient one: $r = 7$ already has the *even* terminal valuation difference

$$V_K^{(J)} - V_I^{(J)} - v = 2 - 3 - 1 = -2$$

shared by every obstruction. Yet it fails $X_{\text{end}} = 1$, its terminal d_J coming out wrong: by the parity lemma below the even difference is necessary for every obstruction, but $r = 7$ shows it is not sufficient.

2.6. **The parity lemma.** The central algebraic fact of this paper is the following.

Lemma 2.5 (Parity lemma). *Suppose the parallel reduction at (r, L) terminates with $X_{\mathrm{end}}(r, L) = 1$, and write $v := v_2(r-1)$. Then*

$$\delta_J := V_K^{(J)} - V_I^{(J)} - v$$

is an even *integer (possibly negative) with $\delta_J \in [-(L-1),\, L-1-v]$.*

Proof. Iterating the update rule from Lemma 2.1 with $c_0 = 2^{-v}$:

$$c_j = 2^{-v} \cdot \prod_{i<j} 2^{v_K^{(i)} - v_I^{(i)}} = 2^{V_K^{(j)} - V_I^{(j)} - v} = 2^{\delta_j}.$$

From $X_J = 3d_J + c_J = 1$, $d_J = (1 - c_J)/3 = (1 - 2^{\delta_J})/3$. The affine relation at index J becomes

$$a_I^{(J)} = 2^{\delta_J} a_K^{(J)} + \frac{1 - 2^{\delta_J}}{3} \qquad \text{in } \mathbb{Q}.$$

Multiplying by 3:

$$3a_I^{(J)} = 3 \cdot 2^{\delta_J} a_K^{(J)} + 1 - 2^{\delta_J}. \tag{1}$$

We split by sign of δ_J. Write $\delta := \delta_J$ throughout.

Case $\delta \geq 0$. All terms in (1) are integers. Reducing modulo 3: $3a_I^{(J)} \equiv 0$ and $3 \cdot 2^{\delta} a_K^{(J)} \equiv 0$, so $1 - 2^{\delta} \equiv 0 \pmod 3$, i.e. $2^{\delta} \equiv 1 \pmod 3$. Since $2 \equiv -1 \pmod 3$, this forces δ even.

Case $\delta < 0$. Write $\delta = -e$, $e > 0$. Multiplying (1) by 2^e:

$$3 \cdot 2^e\, a_I^{(J)} = 3\, a_K^{(J)} + 2^e - 1.$$

All terms are integers; reducing modulo 3 yields $2^e - 1 \equiv 0 \pmod 3$, hence e even, hence δ even.

In both cases δ is even. For the bound, each firing step satisfies $v_K^{(j)} < L - V_K^{(j)}$ and $v_I^{(j)} < L - v - V_I^{(j)}$ (else the respective stop would have fired at j), so

$$V_K^{(J)} \leq L - 1, \qquad V_I^{(J)} \leq L - 1 - v,$$

hence $\delta_J = V_K^{(J)} - V_I^{(J)} - v \in [-(L-1), L-1-v]$. □

2.7. **The shift-s normal form.** Writing $\delta_J = 2s$ with $s \in \mathbb{Z}$, the parity lemma yields

$$c_J = 4^s, \qquad d_J = \frac{1 - 4^s}{3}.$$

We call this the *shift-s normal form* and s the *shift index* of r at level L, denoted $s(r, L)$. In particular, $V_K^{(J)} - V_I^{(J)} = 2s + v$. The obstruction family partitions:

$$\mathcal{O}_L = \bigsqcup_{s \in \mathbb{Z}} \mathcal{O}_L^{G_s}, \qquad \mathcal{O}_L^{G_s} := \{\, r \in \mathcal{O}_L : s(r, L) = s \,\}.$$

For $s=0$ we have $(c_J,d_J)=(1,0)$, meaning $a_I^{(J)}=a_K^{(J)}$. By the bound $\delta_J\in[-(L-1),L-1-v]$ from Lemma 2.5,

$$s=\delta_J/2\in\left[-\tfrac{L-1}{2},\,\tfrac{L-1-v}{2}\right],\qquad\text{so}\qquad |s|\le\left\lfloor\tfrac{L-1}{2}\right\rfloor$$

(using that s is an integer).

2.8. **Simultaneous stop and stop types.**

Corollary 2.6 (Simultaneous stop)**.** *If $X_{\mathrm{end}}(r,L)=1$, then the K-track and I-track satisfy the termination condition $v_\alpha\ge v_2(\beta)$ at the same index J. Moreover, the terminal step valuations satisfy $v_I^{(J)}=v_K^{(J)}+2s$, where s is the shift index of r at level L from the shift-s normal form above.*

Proof. From the shift-s normal form, $a_I^{(J)}=4^s a_K^{(J)}+(1-4^s)/3$, so $A_I^{(J)}=4^sA_K^{(J)}$ and hence $v_I^{(J)}=v_K^{(J)}+2s$. Combining with the residual valuations from Section 2, $v_2(b_I^{(J)})=(L-v)-V_I^{(J)}$ and $v_2(b_K^{(J)})=L-V_K^{(J)}$, and using $V_K-V_I=2s+v$:

$$\begin{aligned}
v_I^{(J)}\ge v_2(b_I^{(J)}) &\iff v_K^{(J)}+2s\ge L-v-V_I^{(J)}\\
&\iff v_K^{(J)}\ge L-V_K^{(J)}\\
&\iff v_K^{(J)}\ge v_2(b_K^{(J)}).
\end{aligned}$$

□

Definition 2.7 (Stop type)**.** The *stop type* at (r,L) is *EE* (equality) if $v_K^{(J)}=v_2(b_K^{(J)})$ and $v_I^{(J)}=v_2(b_I^{(J)})$, and *SS* (strict) if $v_K^{(J)}>v_2(b_K^{(J)})$ and $v_I^{(J)}>v_2(b_I^{(J)})$. By Corollary 2.6 these two cases are exhaustive: a K-track equality forces an I-track equality, and a K-track strict inequality forces an I-track strict inequality.

2.9. **The parameter v and the typical case $v=1$.**

The constructions above are stated for general $v:=v_2(r-1)\ge 1$. The typical case is $v=1$ (equivalently $r\equiv 3\pmod 4$); for $r\equiv 1\pmod 4$ the value of v is ≥ 2. In every formula above, setting $v=1$ recovers the original initial pair $(c_0,d_0)=(\tfrac12,-\tfrac12)$ and the parity relation $V_K^{(J)}-V_I^{(J)}\equiv 1\pmod 2$. The shift-$s$ normal form $(c_J,d_J)=(4^s,(1-4^s)/3)$ is independent of v. The subsequent sections work with the general case throughout; explicit calculations use $v=1$ wherever this aids the reader.

The $v\ge 2$ residues contribute a small but non-negligible share of $\mathcal{O}_L$: direct enumeration gives, for $L=16$, $|\mathcal{O}_L|=7556$ in total, of which 6130 have $v=1$ and the remaining 1426 (roughly 19%) have $v\ge 2$. The numbers in the appendix table count all v.

3. The lift theorem

We now prove Theorem 1.1: for every $r_0\in\mathcal{O}_{L_0}$, both lifts $r_+:=r_0$ and $r_-:=r_0+2^{L_0}$ are obstructions at level L_0+1. The two halves are stated and proved separately as Theorems 3.1 and 3.2; together they constitute

Theorem 1.1. Throughout, $s \in \mathbb{Z}$ is the shift index of r_0 at level L_0, and we use the synchronization $v_I^{(J)} = v_K^{(J)} + 2s$ (Corollary 2.6) freely.

3.1. **The r_+ lift.**

Theorem 3.1. *Let $r_0 \in \mathcal{O}_{L_0}^{G_s}$. Then $r_+ := r_0 \in \mathcal{O}_{L_0+1}$.*

Proof. Set $v := v_2(r_0 - 1) \geq 1$. The starting pairs at $(r_+, L_0 + 1)$ are $(r_0, 2^{L_0+1})$ and $(m_0, 2^{L_0+1-v})$ where $m_0 = (r_0 - 1)/2^v$. These differ from the corresponding pairs at (r_0, L_0) only in the modulus, which is doubled. Since each basic step depends only on $a_K^{(j)}, a_I^{(j)}$ and not on the modulus (as long as the latter does not constrain the step), the parallel reduction at $(r_+, L_0 + 1)$ produces identical $v_K^{(j)}, v_I^{(j)}, c_j, d_j$ for $j \leq J$ to those at (r_0, L_0).

Stop type EE at L_0. Here $v_K^{(J)} = L_0 - V_K^{(J)}$ exactly. At level L_0+1, the K-track modulus at index J has 2-adic valuation $L_0+1-V_K^{(J)} = v_K^{(J)}+1 > v_K^{(J)}$, so the step is permitted. Taking one more step, with $c_J = 4^s$, $d_J = (1-4^s)/3$:

$$c_{J+1} = 4^s \cdot 2^{v_K^{(J)} - v_I^{(J)}} = 4^s \cdot 2^{-2s} = 1,$$

$$d_{J+1} = \frac{3(1-4^s)/3 + 4^s - 1}{2^{v_I^{(J)}}} = 0.$$

After this step, $V_K^{(J+1)} = L_0$, so the K-track residual modulus has valuation 1; by the no-further-step fact (Section 2) no further step is possible. Termination occurs at index $J+1$ with $(c_{J+1}, d_{J+1}) = (1, 0)$, hence $X_{\text{end}} = 1$ and $r_+ \in \mathcal{O}_{L_0+1}^{G_0}$.

Stop type SS at L_0. Here $v_K^{(J)} > L_0 - V_K^{(J)}$. Both sides are integers, so $v_K^{(J)} \geq L_0 + 1 - V_K^{(J)}$, which is exactly the termination condition at level $L_0 + 1$. Termination at $(r_+, L_0 + 1)$ occurs at index J with terminal data unchanged, so $r_+ \in \mathcal{O}_{L_0+1}^{G_s}$. $\square$

3.2. **The r_- lift.**

Theorem 3.2. *Let $r_0 \in \mathcal{O}_{L_0}^{G_s}$. Then $r_- := r_0 + 2^{L_0} \in \mathcal{O}_{L_0+1}$.*

Proof. Write $a_{K,+}^{(j)}, a_{I,+}^{(j)}$ for the values at $(r_+, L_0 + 1)$ (from Theorem 3.1) and $a_{K,-}^{(j)}, a_{I,-}^{(j)}$ for those at $(r_-, L_0 + 1)$; a superscript $-$ on a step valuation (e.g. $v_K^{(J),-}$ below) likewise marks that valuation in the $(r_-, L_0 + 1)$ reduction. The discrepancies

$$\Delta_K^{(j)} := a_{K,-}^{(j)} - a_{K,+}^{(j)}, \qquad \Delta_I^{(j)} := a_{I,-}^{(j)} - a_{I,+}^{(j)}$$

satisfy $\Delta_K^{(0)} = 2^{L_0}$ and $\Delta_I^{(0)} = 2^{L_0 - v}$; the I-track value is

$$\Delta_I^{(0)} = (r_- - 1)/2^v - m_0 = 2^{L_0}/2^v = 2^{L_0 - v},$$

using $v_2(r_- - 1) = \min(v_2(r_0 - 1), L_0) = v$, the last equality because $L_0 > v$. We claim that as long as $V_K^{(j)} < L_0$ and $V_I^{(j)} < L_0 - v$, the two reductions take identical local steps,

$$v_2\Big(3a_{K,-}^{(j)} - 1\Big) = v_2\Big(3a_{K,+}^{(j)} - 1\Big) = v_K^{(j)} \tag{2}$$

(and analogously for the I-track), and the discrepancies have the closed form

$$\Delta_K^{(j)} = 2^{L_0 - V_K^{(j)}} \cdot 3^j, \qquad \Delta_I^{(j)} = 2^{L_0 - v - V_I^{(j)}} \cdot 3^j. \tag{3}$$

The argument at every index rests on the same decomposition. Writing $A_{K,-}^{(j)} := 3a_{K,-}^{(j)} - 1$ and $A_{I,-}^{(j)} := 3a_{I,-}^{(j)} - 1$, whenever (3) holds at step j we have $v_2(3\Delta_K^{(j)}) = L_0 - V_K^{(j)}$ and

$$A_{K,-}^{(j)} = (3a_{K,+}^{(j)} - 1) + 3\Delta_K^{(j)},$$

so the ultrametric identity gives $v_2(A_{K,-}^{(j)}) = \min(v_K^{(j)}, L_0 - V_K^{(j)})$ when $v_K^{(j)} \neq L_0 - V_K^{(j)}$, and $v_2(A_{K,-}^{(j)}) \geq v_K^{(j)} + 1$ when the two are equal (and analogously on the I-track, with $L_0 - v - V_I^{(j)}$ in place of $L_0 - V_K^{(j)}$).

We prove (2) and (3) jointly by induction on j in the range $0 \leq j < J$, where J is the termination index of the reduction at (r_0, L_0). The base case $j = 0$ is immediate. For the inductive step, assume (3) at step j and $j < J$. Since the K-track at (r_0, L_0) has not yet terminated, $v_K^{(j)} < L_0 - V_K^{(j)}$, so the decomposition gives $v_2(A_{K,-}^{(j)}) = v_K^{(j)}$, proving (2) at step j. The basic step then yields $\Delta_K^{(j+1)} = (3\Delta_K^{(j)})/2^{v_K^{(j)}} = 2^{L_0 - V_K^{(j+1)}} \cdot 3^{j+1}$; the same calculation on the I-track completes (3) at step $j+1$.

Stop type SS at L_0. At index J of $(r_+, L_0 + 1)$, $v_K^{(J)} > L_0 - V_K^{(J)}$. The decomposition therefore gives $v_K^{(J),-} := v_2(A_{K,-}^{(J)}) = L_0 - V_K^{(J)} < L_0 + 1 - V_K^{(J)}$, so the K-step at $(r_-, L_0 + 1)$ at index J is permitted. On the I-track the decomposition likewise gives $v_I^{(J),-} := L_0 - v - V_I^{(J)}$. Permitting the I-step additionally requires the strict inequality $v_2(\Delta_I^{(J)}) = L_0 - v - V_I^{(J)} < v_I^{(J)}$, which we now verify. Substituting the synchronisation $v_I^{(J)} = v_K^{(J)} + 2s$ of Corollary 2.6 and $V_K^{(J)} - V_I^{(J)} = 2s + v$ turns this into

$$L_0 - v - V_I^{(J)} < v_I^{(J)} \quad \Longleftrightarrow \quad v_K^{(J)} > L_0 - V_K^{(J)},$$

the K-track inequality already in force. The new exponent in the update rule is $v_K^{(J),-} - v_I^{(J),-} = -2s$, so the extra step closes exactly as in the proof of Theorem 3.1:

$$(c_{J+1}, d_{J+1}) = (1, 0).$$

After it $V_K^{(J+1)} = L_0$ and $V_I^{(J+1)} = L_0 - v$, so both moduli have valuation 1 and no further step is possible for odd a_K, a_I; termination occurs at index $J + 1$, hence $r_- \in \mathcal{O}_{L_0+1}^{G_0}$.

Stop type EE at L_0. Here $v_K^{(J)} = L_0 - V_K^{(J)}$, so the decomposition's equal-valuation branch gives $v_2(A_{K,-}^{(J)}) \geq v_K^{(J)} + 1 = L_0 + 1 - V_K^{(J)}$, which is the K-track stop condition at level $L_0 + 1$. The same argument on the I-track gives $v_2(A_{I,-}^{(J)}) \geq L_0 + 1 - v - V_I^{(J)}$, the I-track stop condition at level $L_0 + 1$. Termination therefore occurs at index J with terminal data unchanged, so $r_- \in \mathcal{O}_{L_0+1}^{G_s}$. □

3.3. **The lift symmetry.**

Proposition 3.3 (Lift symmetry)**.** *Let $r_0 \in \mathcal{O}_{L_0}^{G_s}$. The shift indices of the two level-(L_0+1) lifts $r_+ := r_0$ and $r_- := r_0 + 2^{L_0}$ depend only on the stop type of r_0 at L_0 (Definition 2.7) as follows (both lifts at level $L_0 + 1$):*

Stop type of r_0 at L_0	*shift index of r_+*	*shift index of r_-*
EE	0	s
SS	s	0

Proof. The four entries are read off directly from the case-analysis proofs of Theorems 3.1 and 3.2: each case computes the terminal data of the lift and identifies the resulting shift index. □

So lifting an obstruction with shift index $s \neq 0$ always produces one shift-s obstruction and one shift-zero obstruction at the next level; lifting a shift-zero obstruction produces two shift-zero obstructions (Figure 1). This is the symmetry that drives the recurrence developed in Section 6.

Example 3.4 (The lift of $r = 27$)**.** Continuing Example 2.4: $r_0 = 27 \in \mathcal{O}_5^{G_1}$ terminates at $J = 1$ with stop type EE, the two terminal step valuations meeting their moduli exactly ($v_K^{(J)} = v_2(b_K^{(J)}) = 1$, $v_I^{(J)} = v_2(b_I^{(J)}) = 3$). The EE row of Proposition 3.3 then predicts shift index 0 for r_+ and $s = 1$ for r_-. Indeed $r_+ = 27 \in \mathcal{O}_6^{G_0}$, whose extra step at level 6 collapses the terminal data to $(c_J, d_J) = (1, 0)$, and $r_- = 27 + 2^5 = 59 \in \mathcal{O}_6^{G_1}$, whose terminal data $(4, -1)$ is unchanged from level 5. Thus $\mathcal{O}_6 = \{19, 27, 59\}$ resolves into the new atom 19, the shift-zero lift 27, and the shift-one lift 59.

4. Positive density

Proof of Theorem 1.2. By Theorem 1.1, every $r_0 \in \mathcal{O}_{L_0}$ contributes two distinct elements r_0 and $r_0 + 2^{L_0}$ to $\mathcal{O}_{L_0+1}$. Hence

$$|\mathcal{O}_{L_0+1}| \geq 2|\mathcal{O}_{L_0}|, \qquad \text{equivalently} \qquad \frac{|\mathcal{O}_{L_0+1}|}{2^{L_0+1}} \geq \frac{|\mathcal{O}_{L_0}|}{2^{L_0}},$$

so the density ratio is non-decreasing. Since $\mathcal{O}_L \subseteq \{1, \dots, 2^L\}$ the ratio is bounded above by 1. By monotone convergence, $c_W := \lim_{L\to\infty} |\mathcal{O}_L|/2^L$ exists in $[\rho_{L_0}, 1]$ for every $L_0 \geq 5$. Two substitutions make this explicit:

$$|\mathcal{O}_{16}| = 7556 \;\Rightarrow\; c_W \geq 7556/65536 \approx 0.115,$$
$$\mathcal{O}_6 = \{19, 27, 59\} \;\Rightarrow\; c_W \geq 3/64,$$

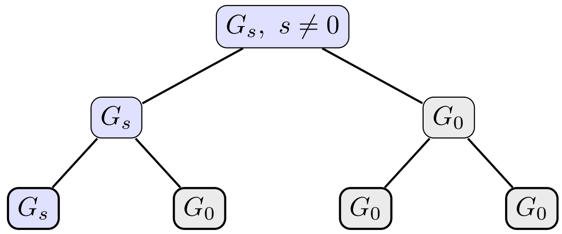

FIGURE 1. Two generations of lifts (Proposition 3.3). A shift-nonzero obstruction (blue) splits into one shift-nonzero and one shift-zero child; a shift-zero obstruction (grey) splits into two shift-zero children. Shift-zero is therefore *absorbing* under lifting: it never produces a shift-nonzero residue. The single persisting shift-nonzero line, shedding one shift-zero child per level, is the combinatorial engine behind the strict lift balance of Corollary 6.4.

the first from the direct enumeration in Appendix A (Table 3), the second the conservative bound at $L_0 = 6$ (Section 2). □

The conservative bound $\rho_6 = 3/64$ uses only the three explicit obstructions $\mathcal{O}_6 = \{19, 27, 59\}$. By raising L_0 and computing $|\mathcal{O}_{L_0}|$ directly, tighter rigorous bounds for c_W follow (Table 1):

L_0	$\lvert\mathcal{O}_{L_0}\rvert$	ρ_{L_0}	rigorous lower bound for c_W
6	3	≈ 0.047	$3/64$
10	91	≈ 0.089	$91/1024$
14	1776	≈ 0.108	$1776/16384$
16	7556	≈ 0.115	$7556/65536$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
32	631538769	≈ 0.147	$631538769/2^{32}$

TABLE 1. Rigorous lower bounds for c_W obtained from finite-level enumeration of $\mathcal{O}_{L_0}$. The ratio $\rho_{L_0} := |\mathcal{O}_{L_0}|/2^{L_0}$ is non-decreasing in L_0 (Theorem 1.2).

These are not asymptotic estimates: they are finite-level computations producing rigorous bounds at the cost of $O(L_0 \cdot 2^{L_0})$ operations to enumerate obstructions modulo 2^{L_0}. The exact value of c_W is not pinned down by this approach; we return to it in Section 6.

5. Full factor complexity

5.1. **Setup.** To each $r \in \mathcal{O}_L$ we associate its binary representation as an L-bit word, least significant bit first:

$$w_r \in \{0,1\}^L, \qquad w_r[k] := \lfloor r/2^k \rfloor \bmod 2, \quad k = 0, \dots, L-1.$$

In particular $w_r[0] = 1$ (since r is odd) and $w_r[L-1]$ is the leading bit of r in this L-bit representation. The *language of obstruction words* at level L is $\mathcal{F}_L := \{w_r : r \in \mathcal{O}_L\}$, and the *factor language* of the obstruction family is

$$\mathcal{F}(\mathcal{O}) := \{\, u \in \{0,1\}^* : u \text{ is a factor of } w_r \text{ for some } r \in \mathcal{O}_L,\ L \geq 5 \,\}.$$

The *factor complexity function* is $p_W(\ell) := |\mathcal{F}(\mathcal{O}) \cap \{0,1\}^\ell|$. Trivially $p_W(\ell) \leq 2^\ell$.

5.2. **The main theorem.**

Theorem (restating Theorem 1.3)**.** *For every $\ell \geq 0$, $p_W(\ell) = 2^\ell$.*

Proof. The case $\ell = 0$ is the empty word, which is trivially a factor of every w_r, so $p_W(0) = 1 = 2^0$. Fix $\ell \geq 1$ and $u = (u_0, u_1, \ldots, u_{\ell-1}) \in \{0,1\}^\ell$. Choose any anchor $r_0 \in \mathcal{O}_6 = \{19, 27, 59\}$ (the set is confirmed by direct enumeration; see Appendix A). Define inductively

$$r_k := r_{k-1} + u_{k-1} \cdot 2^{5+k}, \qquad k = 1, 2, \ldots, \ell.$$

We claim $r_k \in \mathcal{O}_{6+k}$ for each k. The base $k = 0$ is by choice of r_0. For the inductive step from $k-1$ to k, split on the bit u_{k-1}. If $u_{k-1} = 0$ then $r_k = r_{k-1}$ is the r_+ lift, so $r_k \in \mathcal{O}_{6+k}$ by Theorem 3.1. If $u_{k-1} = 1$ then $r_k = r_{k-1} + 2^{5+k}$ is the r_- lift, so $r_k \in \mathcal{O}_{6+k}$ by Theorem 3.2.

By construction,

$$w_{r_\ell}[5+k] = u_{k-1} \qquad (k = 1, \ldots, \ell),$$

so u appears as a factor of w_{r_ℓ} occupying bit positions $5+1, \ldots, 5+\ell$. Thus $u \in \mathcal{F}(\mathcal{O})$ for every $u \in \{0,1\}^\ell$, hence $p_W(\ell) \geq 2^\ell$. Combined with the trivial upper bound, $p_W(\ell) = 2^\ell$. □

In fact each of the three anchors in $\mathcal{O}_6$ individually realises every pattern, as direct enumeration via the lift-symmetry table confirms; the proof above needs only one.

5.3. **Entropy and substitution-shift rigidity.** The *topological entropy* of a language $\mathcal{L} \subseteq \{0,1\}^*$ is $h(\mathcal{L}) := \limsup_{\ell\to\infty} (1/\ell) \log p_{\mathcal{L}}(\ell)$. The factor-complexity function $p_{\mathcal{L}}$ itself goes back to the foundational symbolic-dynamics papers of Morse–Hedlund [MH38, MH40]. See Lind–Marcus [LM95] for general background on entropy of subshifts and subshifts of finite type, Fogg [Fog02] or Allouche–Shallit [AS03] for factor complexity of substitution and automatic sequences, Cassaigne [Cas97] for the general theory of subword complexity in terms of special factors, and Berthé–Rigo [BR10] and Lothaire [Lot02] for modern collected references on combinatorics on words, automata and number theory.

Corollary 5.1. *The topological entropy of $\mathcal{F}(\mathcal{O})$ is* $\log 2$.

This is the maximal value, matching that of the full Bernoulli shift $\{0,1\}^{\mathbb{Z}}$.

Corollary 5.2. *The obstruction family $(\mathcal{O}_L)_{L\geq 5}$, viewed as a graded family of finite languages over $\{0,1\}$, is*

(i) not the language of a proper subshift of finite type, that is, one defined by a non-empty list of forbidden factors;
(ii) not the language of a proper sofic shift in the classical sense (image of a proper SFT under a one-block factor map);
(iii) not the factor language of a primitive substitution shift in the sense of Pansiot [Pan84] *or Devyatov* [Dev15].

Proof. (i) A non-trivial SFT defined by a non-empty finite list of forbidden subwords has entropy strictly less than $\log 2$ (Lind–Marcus [LM95]), contradicting Corollary 5.1. (ii) A non-trivial sofic shift is the image of an SFT under a one-block factor map and inherits the entropy constraint. (iii) Primitive substitution shifts have factor complexity in one of the polynomial classes $\Theta(1)$, $\Theta(n)$, $\Theta(n \log\log n)$, $\Theta(n \log n)$, $\Theta(n^2)$ (Pansiot [Pan84]; refined by Devyatov [Dev15]), and therefore have topological entropy zero. But the obstruction family has $p_W(\ell) = 2^\ell$, hence entropy $\log 2 > 0$; so it cannot be the factor language of a primitive substitution shift. $\square$

Remark 5.3. The restriction to *primitive* substitution shifts in Corollary 5.2(iii) is the scope of the Pansiot–Devyatov classification we invoke. The entropy obstruction of Corollary 5.1 in fact excludes the entire zero-entropy regime, including all linearly recurrent subshifts, which contain the primitive substitution shifts [DHS99]; the substitutive candidates it does *not* reach are therefore the non-primitive shifts of positive entropy. Whether $\mathcal{F}(\mathcal{O})$ can be realised as the factor language of such a shift remains open (see Section 8). The same question can be posed within the S-adic framework [BD14], which builds sequences by composing morphisms from a possibly infinite set and so reaches beyond single primitive substitutions.

6. The G_s classification

6.1. **Atomic obstructions.** An obstruction $r \in \mathcal{O}_L$ is *atomic at level L* if $r \bmod 2^{L-1} \notin \mathcal{O}_{L-1}$. We write $\mathcal{A}_L \subseteq \mathcal{O}_L$ and $\mathcal{A}_L^{G_s} := \mathcal{A}_L \cap \mathcal{O}_L^{G_s}$.

Lemma 6.1 (Atom decomposition)**.** *For every $L \geq 5$,*

$$|\mathcal{O}_L| = \sum_{L_0=5}^{L} |\mathcal{A}_{L_0}| \cdot 2^{L-L_0}.$$

Proof. We show that every $r \in \mathcal{O}_L$ has a unique smallest $L_0 \geq 5$ with $r \bmod 2^{L_0} \in \mathcal{A}_{L_0}$. *Existence:* starting from r at level L, consider the descending sequence $r_k := r \bmod 2^{L-k}$ for $k = 0, 1, \ldots, L-5$ — as k increases the level $L-k$ decreases, so larger k gives a coarser residue. By the lift theorem (Theorem 1.1), whenever $r_{k+1} \in \mathcal{O}_{L-k-1}$, the level-$(L-k)$ residue $r_k = r \bmod 2^{L-k}$ is one of the two lifts of r_{k+1}, hence lies in $\mathcal{O}_{L-k}$. The converse need not hold. Let L_0 be the smallest level ≥ 5 for which $r \bmod 2^{L_0} \in \mathcal{O}_{L_0}$. We check that $r \bmod 2^{L_0-1} \notin \mathcal{O}_{L_0-1}$. For $L_0 > 5$ this is the minimality of L_0; for $L_0 = 5$ it follows from $\mathcal{O}_4 = \emptyset$ (Remark 2.3). Hence $r \bmod 2^{L_0} \in \mathcal{A}_{L_0}$. *Uniqueness:* the level L_0 is the smallest such by definition, hence unique.

Given the atomic anchor $r_0 := r \bmod 2^{L_0}$, the residue r is determined by r_0 together with the bits $\{u_{L_0}, \dots, u_{L-1}\}$ encoding the lift choices (Theorems 3.1 and 3.2). Each atomic anchor at level L_0 thus generates 2^{L-L_0} obstructions at level L. □

6.2. No shift-zero atoms.

Theorem 6.2. *For every $L \geq 7$, $|\mathcal{A}_L^{G_0}| = 0$.*

The conclusion also holds at the base levels $L = 5, 6$ (see Corollary 6.3); the bound $L \geq 7$ in the statement is conservative. At $L = 5$ it is vacuous, since $|\mathcal{O}_5^{G_0}| = 0$; at $L = 6$ the unique shift-zero obstruction is $r = 27$, whose reduction terminates at $J = 2$ with $V_K^{(J)} = 5 = L - 1$, so it would fall in Case B below. We discharge both base cases formally by direct enumeration in Corollary 6.3.

Proof. Fix $L \geq 7$ and suppose $r \in \mathcal{O}_L^{G_0}$, so $X_{\mathrm{end}}(r, L) = 1$ with shift index $s = 0$, i.e. $(c_J, d_J) = (1, 0)$. Write $r' := r \bmod 2^{L-1}$. Recall that $v_K^{(j)}, v_I^{(j)}$ are the per-step valuations and $V_K^{(j)}, V_I^{(j)}$ their running sums on the two tracks. We first note $J \geq 2$: at $J = 0$ the terminal pair is $(2^{-v}, -2^{-v})$, whose second coordinate -2^{-v} is nonzero for every v, contradicting $d_J = 0$; at $J = 1$ the update rule gives

$$d_1 = \frac{3 \cdot (-2^{-v}) + 2^{-v} - 1}{2^{v_I^{(0)}}} = -\frac{1 + 2^{1-v}}{2^{v_I^{(0)}}},$$

whose numerator is strictly negative for $v \geq 1$ (since $2^{1-v} + 1 > 0$), so $d_1 \neq 0$, again contradicting $d_J = 0$. Hence $J \geq 2$, so the case distinction below over the indices $J - 1$ and J is well-defined. The parallel reduction at $(r', L - 1)$ — the level-$(L - 1)$ companion of the level-L reduction at (r, L), with the same two tracks K, I — starts from the same residue class $a_K \bmod 2^{L-1}, a_I \bmod 2^{L-1-v}$ as at (r, L), with the modulus reduced by one power of two; the two K-tracks (resp. I-tracks) differ as integers by $\Delta_K^{(j)} = 2^{L-1-V_K^{(j)}} \cdot 3^j$ (resp. $\Delta_I^{(j)} = 2^{L-1-v-V_I^{(j)}} \cdot 3^j$), in direct analogy with the lift-minus calculation of Theorem 3.2. Whenever $v_K^{(j)} < L - 1 - V_K^{(j)}$, the ultrametric identity forces $v_2(3a_K^{(j)} - 1)$ to agree at (r, L) and $(r', L - 1)$. Consequently the per-step valuations $v_K^{(j)}, v_I^{(j)}$ and hence the affine data (c_j, d_j) coincide between the two reductions until the level-$(L - 1)$ stop fires.

Termination of the reduction at (r, L) at index J means at least one track satisfies its stop condition there; by Corollary 2.6, both do. Since $s = 0$, at the terminal index J the synchronisation $v_I^{(J)} = v_K^{(J)} + 2s = v_K^{(J)}$ and $V_K^{(J)} - V_I^{(J)} = v$ holds (Lemma 2.5). Hence the K-track stop condition $v_K^{(J)} \geq L - V_K^{(J)}$ and the I-track stop condition $v_I^{(J)} \geq (L - v) - V_I^{(J)}$ at $j = J$ are equivalent: substituting $V_I^{(J)} = V_K^{(J)} - v$ and $v_I^{(J)} = v_K^{(J)}$,

$$v_I^{(J)} \geq (L - v) - V_I^{(J)} \iff v_K^{(J)} \geq (L - v) - (V_K^{(J)} - v) = L - V_K^{(J)}.$$

Both stop conditions at $j = J$ therefore read

$$V_K^{(J)} + v_K^{(J)} \geq L.$$

This equivalence holds only at the terminal index J: at $j < J$ the K- and I-track stop conditions differ, so the child stops below must be ruled out on each track separately. We split by the K-track value $V_K^{(J)}$ (equivalently $V_K^{(J-1)} + v_K^{(J-1)}$): either $V_K^{(J)} < L-1$ (Case A) or $V_K^{(J)} \geq L-1$ (Case B); these exhaust the integer possibilities.

Claim: no level-$(L-1)$ *stop fires on either track at an index* $J' \leq J-2$. Suppose the K-track stop fired there, so $V_K^{(J')} + v_K^{(J')} \geq L-1$. The (r, L) reduction still takes its J'-th step, since the level-L stop has not yet fired at $J' < J$, so $V_K^{(J'+1)} \geq L-1$. By the no-further-step fact (Section 2) the (r, L) reduction then stops at $J'+1 \leq J-1$, contradicting the definition of J. The same bound on the I-track, $V_I^{(J'+1)} \geq L-1-v$, rules out a child I-stop at $J' \leq J-2$ by the same fact. This proves the claim.

At index $J-1$, a child I-stop requires $V_I^{(J)} \geq L-1-v$. By the terminal synchronisation $V_I^{(J)} = V_K^{(J)} - v$ (from $s = 0$),

$$V_I^{(J)} \geq L-1-v \quad \Longleftrightarrow \quad V_K^{(J)} \geq L-1,$$

the Case B branch below; so in Case A neither child stop fires before index J.

Case A: $V_K^{(J)} < L-1$. The K-track stop at level $L-1$ does not fire at any index $\leq J-1$: the indices $J' \leq J-2$ are ruled out by the preceding paragraph, and $J' = J-1$ would force $V_K^{(J)} = V_K^{(J-1)} + v_K^{(J-1)} \geq L-1$, against the Case-A hypothesis $V_K^{(J)} < L-1$. So the $(r', L-1)$ reduction runs at least as far as index J; at $j = J$,

$$V_K^{(J)} + v_K^{(J)} \geq L > L-1,$$

hence the K-track stop at level $L-1$ fires at J and the reduction terminates there. By the terminal synchronisation already noted the I-track stop also fires at J; for termination only the first track to stop matters. The terminal data of the two reductions coincide: $(c_J, d_J) = (1, 0)$, so $r' \in \mathcal{O}_{L-1}^{G_0}$ and r is not atomic.

Case B: $V_K^{(J)} \geq L-1$, *equivalently* $V_K^{(J-1)} + v_K^{(J-1)} \geq L-1$. The quantity $V_K^{(J)} = V_K^{(J-1)} + v_K^{(J-1)}$ also satisfies $V_K^{(J)} < L$, since the level-L stop does not fire at $J-1$. With $V_K^{(J)} \geq L-1$ this forces the only integer value $V_K^{(J)} = L-1$. Hence the K-track stop at level $L-1$ fires at $J-1$, and the $(r', L-1)$ reduction terminates one step earlier, at index $J-1$, with terminal data (c_{J-1}, d_{J-1}) (whether the I-track stop at level $L-1$ also fires at $J-1$ is immaterial for termination). From the update rule (Lemma 2.1) at index $J-1$,

$$c_J = c_{J-1} \cdot 2^{v_K^{(J-1)} - v_I^{(J-1)}}, \qquad d_J = \frac{3d_{J-1} + c_{J-1} - 1}{2^{v_I^{(J-1)}}}.$$

Setting $c_J = 1$, $d_J = 0$ and reading off the numerator of d_J: $3d_{J-1}+c_{J-1}-1 = 0$, i.e. $X_{J-1} = 1$. By the parity lemma applied at the terminal index $J-1$ of the reduction at $(r', L-1)$, the terminal pair is in shift-s' normal form for some $s' \in \mathbb{Z}$, so $r' \in \mathcal{O}_{L-1}^{G_{s'}}$.

In either case, $r' \in \mathcal{O}_{L-1}$, so r is not atomic. □

Corollary 6.3. *For every $L \geq 5$, $|\mathcal{A}_L^{G_0}| = 0$.*

Proof. For $L \geq 7$, this is Theorem 6.2. At the two base levels direct enumeration (verified by `count_obstructions.py`) gives $\mathcal{A}_5 = \{27\}$ and $\mathcal{A}_6 = \{19\}$, each with shift index $s = 1 \neq 0$, so $|\mathcal{A}_5^{G_0}| = |\mathcal{A}_6^{G_0}| = 0$ as well. □

6.3. **Strict lift balance and series representation.** Write $g_L := |\mathcal{O}_L^{G_0}|$ and $h_L := |\mathcal{O}_L^{G_{\neq 0}}|$. The lift symmetry of Proposition 3.3 gives, per parent at $L-1$ (Table 2):

Parent at $L-1$	Contribution to g_L	Contribution to h_L
G_0	2	0
G_s, $s \neq 0$	1	1

Table 2. Per-parent contribution of a level-$(L-1)$ obstruction to the level-L shift-zero and shift-nonzero populations under the two lifts $r_\pm$.

Summing and adding the atomic contributions, which by Corollary 6.3 occur only in h_L:

Corollary 6.4 (Strict lift balance)**.** *For every $L \geq 6$,*

$$g_L = 2g_{L-1} + h_{L-1}, \qquad h_L = h_{L-1} + |\mathcal{A}_L^{G_{\neq 0}}|.$$

Adding, $|\mathcal{O}_L| = 2|\mathcal{O}_{L-1}| + |\mathcal{A}_L^{G_{\neq 0}}|$.

Theorem 6.5. *The density constant c_W admits the convergent series representation*

$$c_W = \frac{3}{64} + \sum_{L \geq 7} \frac{|\mathcal{A}_L^{G_{\neq 0}}|}{2^L},$$

with non-negative terms. In particular, $|\mathcal{A}_L^{G_{\neq 0}}|/2^L \to 0$ as $L \to \infty$.

Proof. By Lemma 6.1 every $r \in \mathcal{O}_L$ has a unique atomic anchor at some level $L_0 \in \{5, 6, \dots, L\}$. By Corollary 6.3, $|\mathcal{A}_{L_0}^{G_0}| = 0$ for every $L_0 \geq 5$, so each atom contributes to $\mathcal{A}_{L_0}^{G_{\neq 0}}$. Direct enumeration at the two base levels (verified by `count_obstructions.py`) gives $\mathcal{A}_5 = \{27\}$ and $\mathcal{A}_6 = \{19\}$, both with shift index $s = 1$, so $|\mathcal{A}_5^{G_{\neq 0}}| = |\mathcal{A}_6^{G_{\neq 0}}| = 1$. Hence for every $L \geq 6$,

$$|\mathcal{O}_L| = 2^{L-5} + 2^{L-6} + \sum_{L_0=7}^{L} |\mathcal{A}_{L_0}^{G_{\neq 0}}| \cdot 2^{L-L_0} = 3 \cdot 2^{L-6} + \sum_{L_0=7}^{L} |\mathcal{A}_{L_0}^{G_{\neq 0}}| \cdot 2^{L-L_0}.$$

Dividing both sides by 2^L,

$$\rho_L = \frac{|\mathcal{O}_L|}{2^L} = \frac{3}{64} + \sum_{L_0=7}^{L} \frac{|\mathcal{A}_{L_0}^{G_{\neq 0}}|}{2^{L_0}}.$$

By Theorem 1.2, ρ_L is non-decreasing in L and bounded above by 1, so its limit c_W exists in $(0,1]$. The right-hand side is the L-th partial sum of a non-negative series with fixed first term $3/64$; since the partial sums converge to c_W, the series converges, with sum c_W and tail $|\mathcal{A}_L^{G_{\neq 0}}|/2^L \to 0$ as $L \to \infty$. □

6.4. **The shift-index window.** The G_s classification raises the question of which shift classes can be nonempty at a given level. We show that the *nontrivial* classes ($|s| \geq 2$) occupy only a window of shift indices of width $O(L)$. The engine is a lower bound on the termination index J in terms of $|s|$: an obstruction of large shift index reduces slowly. The bound is sharp — for instance $r = 15877 \in \mathcal{O}_{15}^{G_2}$ terminates at $J = 7 = 2 \cdot 2 + 3$ with terminal data $(c_J, d_J) = (16, -5) = \big(4^2, (1-4^2)/3\big)$, and $r = 1015979 \in \mathcal{O}_{21}^{G_{-2}}$ likewise has $J = 7$. We emphasise at the outset what the resulting window does *not* give: it bounds the *number* of nonempty shift classes per level, not their populations, and so does not by itself resolve the growth question of Section 8.1 (Remark 6.15).

Notation. Throughout this subsection $s = s(r, L)$ is the (signed) shift index of an obstruction $r \in \mathcal{O}_L$, $J = J(r, L)$ its termination index, $v = v_2(r-1)$, and

$$q := V_I^{(J)} - J.$$

Since the I-track takes J steps, each of valuation ≥ 1 (as $3a_I^{(j)} - 1$ is even for odd $a_I^{(j)}$), $V_I^{(J)} \geq J$, so $q \geq 0$.

Lemma 6.6 (K-track prefix)**.** *For every odd $r > 1$ with $v = v_2(r-1)$, the K-track of the parallel reduction satisfies $v_K^{(j)} = 1$ for $0 \leq j \leq v-2$ and $v_K^{(v-1)} \geq 2$. Hence $V_K^{(j)} = j$ for $0 \leq j \leq \min(v-1, J)$.*

Proof. Write $r = 1 + 2^v m$ with m odd. We claim $a_K^{(j)} = 1 + 3^j 2^{v-j} m$ for $0 \leq j \leq v-1$. The base case $j = 0$ is the definition. If it holds at $j \leq v-2$, then $3a_K^{(j)} - 1 = 2 + 3^{j+1} 2^{v-j} m = 2\big(1 + 3^{j+1} 2^{v-j-1} m\big)$, and $1 + 3^{j+1} 2^{v-j-1} m$ is odd because $v - j - 1 \geq 1$; thus $v_K^{(j)} = 1$ and $a_K^{(j+1)} = 1 + 3^{j+1} 2^{v-(j+1)} m$, completing the induction. At $j = v-1$, $3a_K^{(v-1)} - 1 = 2 + 3^v 2m = 2\big(1 + 3^v m\big)$ with $1 + 3^v m$ even (both 3^v and m are odd), so $v_K^{(v-1)} \geq 2$. The cumulative statement follows by summing the unit valuations. (For $v = 1$ the first claim is vacuous and the second reads $v_K^{(0)} \geq 2$.) □

Lemma 6.7 (Minimal depth)**.** *Every obstruction with shift index $s \geq 0$ has $J \geq v$.*

Proof. If $J \le v-1$, then $V_K^{(J)} = J$ by Lemma 6.6. But also $V_K^{(J)} = V_I^{(J)} + 2s + v \ge J + v$, using $V_I^{(J)} \ge J$ and $s \ge 0$. Combined with $V_K^{(J)} = J$ this forces $v \le 0$, a contradiction. ☐

Lemma 6.8 (Sign-free obstruction identity). *For every $r \in \mathcal{O}_L$, with cumulative valuations along the two tracks and termination index J,*

$$\sum_{j=0}^{J} 3^{\,J-j} 2^{V_K^{(j)}} \;=\; 3^{\,J+1} \;+\; 2^v \sum_{j=0}^{J} 3^{\,J-j} 2^{V_I^{(j)}}. \tag{4}$$

The sign of s does not enter; equivalently, with the terminal terms removed,

$$\sum_{j=0}^{J-1} 3^{\,J-j}\big(2^{V_K^{(j)}} - 2^{V_I^{(j)}+v}\big) \;=\; 3^{\,J+1} + 2^{V_I^{(J)}+v}\big(1-4^s\big). \tag{5}$$

Proof. Set $\tilde n_j := 2^{V_I^{(j)}+v} d_j$. Iterating $c_{j+1} = c_j 2^{v_K^{(j)} - v_I^{(j)}}$ from $c_0 = 2^{-v}$ (Lemma 2.1) gives $c_j = 2^{V_K^{(j)} - V_I^{(j)} - v}$, so $2^{V_I^{(j)}+v} c_j = 2^{V_K^{(j)}}$. The update $d_{j+1} = (3d_j + c_j - 1)/2^{v_I^{(j)}}$ then yields

$$\tilde n_{j+1} = 2^{V_I^{(j)}+v}\big(3d_j + c_j - 1\big) = 3\,\tilde n_j + 2^{V_K^{(j)}} - 2^{V_I^{(j)}+v}, \qquad \tilde n_0 = 2^v d_0 = -1.$$

Unrolling and multiplying by 3, using $3\tilde n_J = 2^{V_I^{(J)}+v}\cdot 3d_J = 2^{V_I^{(J)}+v}(1-4^s)$ from the shift-s normal form, gives (5). Since $V_K^{(J)} = V_I^{(J)} + v + 2s$, the right-hand side of (5) equals

$$3^{J+1} + 2^{V_I^{(J)}+v} - 2^{V_K^{(J)}}.$$

Moving the two terminal terms into the respective sums — extending each from $j = J-1$ to $j = J$ — turns (5) into (4). ☐

The bound itself splits by the size and sign of s. The two large-shift regimes are analytic; the small-shift tails are finite computations.

Lemma 6.9 (Large negative shift). *Every obstruction with $s \le -4$ has $J \ge 2|s| + 3$.*

Proof. Write $a = |s|$ and abbreviate

$$\hat A_K = \sum_{j=0}^{J} 3^{J-j} 2^{V_K^{(j)}}, \qquad \hat A_I = \sum_{j=0}^{J} 3^{J-j} 2^{V_I^{(j)}},$$

so (4) reads $\hat A_K = 3^{J+1} + 2^v \hat A_I$. We use two elementary bounds. Keeping only the $j = J$ term of $\hat A_I$ (all terms positive) and using $V_I^{(J)} + v = V_K^{(J)} + 2a$,

$$2^v \hat A_I \;\ge\; 2^{V_I^{(J)}+v} \;=\; 2^{V_K^{(J)}+2a}.$$

Since the valuations increase by at least 1 per step, $V_K^{(j)} \le V_K^{(J)} - (J-j)$, whence

$$\hat{A}_K \;\le\; 2^{V_K^{(J)}} \sum_{i=0}^{J} (3/2)^i \;<\; 2^{V_K^{(J)}} \cdot \frac{3^{J+1}}{2^J}.$$

As $\hat{A}_K > 2^v \hat{A}_I$, combining gives $2^{V_K^{(J)}+2a} < 2^{V_K^{(J)}} 3^{J+1}/2^J$, i.e.

$$3^{\,J+1} > 2^{\,J+2a}. \tag{N}$$

The ratio $f(J) := 3^{J+1}/2^{J+2a}$ is strictly increasing, and

$$f(2a+2) = \frac{3^{\,2a+3}}{2^{\,4a+2}} = \frac{27}{4}\Big(\frac{9}{16}\Big)^a < 1 \iff a \ge 4.$$

Hence for $a \ge 4$ every $J \le 2a+2$ violates (N), so $J \ge 2a+3$. □

Lemma 6.10 (Large positive shift)**.** *Every obstruction with $s \ge 4$ has $J \ge 2s+3$.*

Proof. Recall $q = V_I^{(J)} - J \ge 0$, $v = v_2(r-1)$, and $s \ge 4$ here. By Lemma 6.7, $v \le J$. Write the O”-form (5) as $A_K - 2^v A_I = 3^{J+1} + 2^{\,J+q+v}(1-4^s)$, where $A_K = \sum_{j=0}^{J-1} 3^{J-j} 2^{V_K^{(j)}}$ and A_I is the analogous I-sum; here $V_I^{(J)} = J+q$ has been substituted into the terminal exponent. The two sums range over the structural families (v bounds the forced K-prefix, q the terminal I-excess): by Lemma 6.6 the K-valuations satisfy $V_K^{(j)} = j$ for $j \le v-1$, $V_K^{(j)} \ge j+1$ for $v \le j \le J-1$; the I-valuations satisfy $V_I^{(0)} = 0$, $V_I^{(j+1)} > V_I^{(j)}$, $V_I^{(J)} = J+q$. Minimising A_K and maximising A_I over these families *independently* can only decrease $A_K - 2^v A_I$, so the decoupled deficit

$$D := (A_K^{\min} - 2^v A_I^{\max}) - \big(3^{J+1} + 2^{J+q+v}(1-4^s)\big)$$

satisfies $D > 0 \Rightarrow$ no obstruction with these parameters. Both optima are attained by saturating the structural bounds just listed: since the common weight 3^{J-j} in A_K and A_I decreases strictly in j, each sum is extremised at the envelope of its constraints. Concretely, $A_I^{\max}$ puts all I-excess in the first step (I-valuations as large as the endpoint $V_I^{(J)} = J+q$ and strict monotonicity allow), while $A_K^{\min}$ keeps the K-valuations on their lower envelope, deferring the single forced jump of Lemma 6.6 to the final, uncounted step. Both extremal sums are independently confirmed by a brute optimisation over the families in Appendix A. This gives the closed forms

$$A_I^{\max} = 3^J + 2^q\big(2\cdot 3^J - 3\cdot 2^J\big), \qquad A_K^{\min} = 3^{J+1} + 2^v 3^{\,J-v+1} - 6\cdot 2^J,$$

and, after the 3^{J+1} cancels,

$$D = C_0(J,v) + 2^{\,v+q}\,\Gamma(J,s),$$

where $\Gamma(J,s) = 2^J(4^s+2) - 2\cdot 3^J$ and $C_0(J,v) = 2^v 3^J\big(3^{1-v} - 1\big) - 6\cdot 2^J$. Here Γ governs the q-dependence of D and C_0 the q-independent (J,v)-terms; the $q=0$ specialisation below splits D further into a $q=0$ envelope Φ and a v-correction R. For $s \ge 3$ and $1 \le J \le 2s+2$ one has $\Gamma(J,s) > 0$: the sign change $2^J(4^s+2) = 2\cdot 3^J$ lies at $J \approx s\log_{3/2} 4$, and $s\log_{3/2} 4 > 2s+2$ for $s \geq 3$ places it past the window. Hence D is non-decreasing in q and $q=0$ is the worst case. At $q=0$, $D(J,s,v,0) = 2^v\,\Phi(J,s) + R(J,v)$, where $\Phi(J,s) = 2^J(4^s+2) - 3^{J+1}$ and $R(J,v) = 2^v 3^{\,J-v+1} - 6\cdot 2^J$. For $s \geq 4$ and $J \leq 2s+2$, $\Phi(J,s) > 3$ (worst case $J = 2s+2$: $\Phi(2s+2,s) > 3 \iff (16/9)^s > 27/4$, which holds for $s \geq 4$). Finally $R(J,v) = 3\cdot 2^v\bigl(3^{\,J-v} - 2^{\,J-v+1}\bigr)$. When $J-v \geq 2$, $R \geq 0$, so $D(J,s,v,0) = 2^v\Phi + R \geq 2^v\Phi > 0$. When $J - v \in \{0,1\}$, $R = -3\cdot 2^v$, so $D(J,s,v,0) = 2^v(\Phi-3) > 0$ since $\Phi > 3$. Thus $D > 0$ for all $q \geq 0$ and all $1 \leq v \leq J \leq 2s+2$, so no obstruction has $J \leq 2s+2$. $\square$

The remaining strata are the small shifts $|s| \in \{2,3\}$. After the analytic bounds only finitely many configurations survive, and they are settled by an exact finite computation. The key device is a bounded carry automaton, which decides a whole stratum for *all* q at once.

Lemma 6.11 (Bounded carry automaton)**.** *Fix a stratum (s,J) and v. Reading the integer identity (4) bit position by bit position defines a finite automaton whose state is (carry, $\#\{V_K$ placed$\}$, $\#\{V_I$ placed$\}$). Each strictly increasing valuation contributes one set bit. The carry stays bounded, each transition halving it towards a fixed invariant interval. An obstruction with parameter q then corresponds exactly to an accepting run (carry 0, all valuations placed). The positions factor as prefix → block A → block B → tail, where block A carries a single q- and position-independent transition T_A whose* length *is the only q-dependence, block B has fixed width $2|s|+1$, and the tail accepts iff the incoming carry is 0 (Figure 2). Since the state space is finite, the reachable-set sequence $R_m = T_A^m(R_0)$ is eventually periodic; acceptance therefore depends on q only through $R_{m(q)}$ and is periodic in q past a preperiod. A finite check over one preperiod plus one period decides the stratum for every $q \geq 0$.*

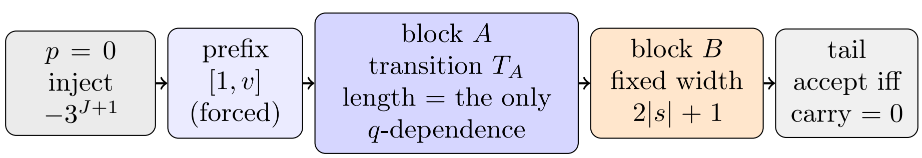

FIGURE 2. Block factorisation of the carry automaton of Lemma 6.11. Only the *length* of block A depends on q; the transition T_A does not. As $T_A^m(R_0)$ is eventually periodic, a finite computation over one preperiod and one period decides the stratum for all q. The fixed-point data (preperiods, periods, reachable-state counts) per stratum are tabulated in Appendix A.

Lemma 6.12 (Small-shift tails)**.** *No obstruction has $|s| \in \{2,3\}$ and $J \leq 2|s|+2$.*

Proof. For $s \le -2$ the inequality (N) (which used only $|s| \ge 2$) rules out every $J \le 2|s|+2$ except the strata $(|s|, J) \in \{(2,5),(2,6),(3,8)\}$; each is closed for all q by Lemma 6.11. For $s \ge 2$ the decoupled deficit D of Lemma 6.10 is positive except on a finite, explicitly enumerated set: for $s = 3$ only the cells ($J = 8$, $q = 0$, $v \in \{2,\dots,8\}$) survive, and for $s = 2$ only ($J = 5$, $q \in \{0,1\}$) together with the single stratum $J = 6$. The stratum ($s = 2, J = 6$) is special: there $\Gamma(6,2) < 0$ leaves q unbounded, and it is closed for all q by Lemma 6.11. The finitely many residual cells are closed by exhaustive search over the coupled valuation sequences. All these checks are reproducible and are recorded in Appendix A. ☐

Theorem 6.13 (Depth bound)**.** *Every obstruction with* $|s| \ge 2$ *satisfies* $J \ge 2|s|+3$.

Proof. For $s \ge 4$ this is Lemma 6.10; for $s \le -4$, Lemma 6.9; for $|s| \in \{2,3\}$, Lemma 6.12 rules out $J \le 2|s|+2$. ☐

Corollary 6.14 (Shift-index window)**.** *Let* $r \in \mathcal{O}_L$ *with* $v = v_2(r-1)$*. If* $s \ge 2$*, then*

$$s \le \frac{L-4-v}{4} \le \left\lfloor \tfrac{L-5}{4} \right\rfloor;$$

if $s \le -2$*, then* $s \ge -\dfrac{L-4-v}{2}$*. In particular at most* $O(L)$ *of the classes* $\mathcal{O}_L^{G_s}$ *are nonempty.*

Proof. For $s \ge 2$, Theorem 6.13 gives $J \ge 2s+3$, and $V_K^{(J)} = V_I^{(J)} + 2s + v \ge J + 2s + v \ge 4s + 3 + v$; with $V_K^{(J)} \le L-1$ (Lemma 2.5) this yields $4s+3+v \le L-1$, i.e. $s \le (L-4-v)/4 \le (L-5)/4$ as $v \ge 1$. For $s \le -2$, $V_I^{(J)} \ge J \ge 2|s|+3$ and $V_I^{(J)} \le L-1-v$ (Lemma 2.5) give $2|s|+3 \le L-1-v$, i.e. $s \ge -(L-4-v)/2$. The shift indices of the nontrivial classes thus lie in an interval of length $O(L)$; the three classes $|s| \le 1$ add a constant. ☐

The window is asymmetric: the positive side scales like $L/4$, the negative side like $L/2$. The asymmetry is structural — on the positive side $V_K^{(J)} = V_I^{(J)} + 2s + v$ contributes the extra $2s$ gap that yields the factor 4, whereas on the negative side the high-ending track is the I-track and only the factor 2 survives. The depth bound $J \ge 2|s|+3$ of Theorem 6.13, by contrast, is fully two-sided.

Remark 6.15. Corollary 6.14 bounds the *number* of nonempty shift classes at level L by $O(L)$, hence the support of the shift distribution is a window of width $O(L)$. It does not bound the *population* $|\mathcal{O}_L^{G_s}|$ of any class, and in particular gives no bound on $|\mathcal{A}_L^{G\neq 0}|$; it therefore does not resolve the growth question of c_W taken up in Section 8.1.

6.5. **The remaining open question.** Theorem 6.5 reduces the question of c_W's exact value to controlling the growth of $|\mathcal{A}_L^{G\neq 0}|$. A bound of the form $|\mathcal{A}_L^{G\neq 0}| \leq C \cdot \alpha^L$ with $\alpha < 2$ would give a geometrically convergent series with explicit error bounds. We do not establish such a bound here; we return to the question in Section 8.

7. The T_+ side

The classical Collatz map T_+ admits the same obstruction theory. We transfer the framework of Section 2 to T_+ and show that the residue involution $r \mapsto \overline{r} := 2^L - r$ realises an explicit bijection between the two obstruction families, step-by-step at the level of the parallel reduction.

7.1. **The T_+ parallel reduction.** Fix a positive odd integer r with $v_2(r+1) \geq 1$, and write $v := v_2(r+1)$ and $m := (r+1)/2^v$; then m is odd. For a level $L > v$, run the parallel reduction with T_+:

- the *K-track*, starting from $(a_K^{(0)}, b_K^{(0)}) := (r, 2^L)$ with basic step $a_K' = (3a_K+1)/2^{v_K}$, $v_K := v_2(3a_K+1)$;
- the *I-track*, starting from $(a_I^{(0)}, b_I^{(0)}) := (m, 2^{L-v})$ with the analogous basic step.

The initial relation $m = (r+1)/2^v$ writes as $a_I^{(0)} = 2^{-v} a_K^{(0)} + 2^{-v}$, so the affine relation $a_I^{(j)} = c_j a_K^{(j)} + d_j$ holds at index 0 with $c_0 = 2^{-v}$, $d_0 = +2^{-v}$. The derivation of Lemma 2.1, applied with T_+ in place of T_- and using $3a_I + 1 = c_j(3a_K+1) + (3d_j - c_j + 1)$, yields the update rule

$$c_{j+1} = c_j \cdot 2^{v_K^{(j)} - v_I^{(j)}}, \qquad d_{j+1} = \frac{3d_j - c_j + 1}{2^{v_I^{(j)}}}.$$

(The numerator changes sign relative to the T_- rule.) We define the T_+-analogue of the X-invariant by

$$X_{\text{end}}^+(r, L) := c_J - 3d_J,$$

where J is the termination index. An odd r with $v_2(r+1) \geq 1$ and $L > v_2(r+1)$ is a *T_+-obstruction residue at level L* if $X_{\text{end}}^+(r, L) = 1$; we write $\mathcal{O}_L^+ \subseteq \{1, \ldots, 2^L\}$ for the set of such residues.

Remark 7.1. The sign convention $X_{\text{end}}^+ := c - 3d$ (rather than $3d + c$) is the one forced by the affine algebra: with this combination the analogue of the parity lemma (Lemma 2.5) yields the shift-s normal form $(c_J, d_J) = (4^s, (4^s-1)/3)$, and the residue-level bijection of Theorem 7.3 below is exact.

7.2. **The residue involution.**

Lemma 7.2 (Residue-level involution)**.** *Let r be the T_- residue, odd with $v := v_2(r-1) \geq 1$ (the T_- convention, not the T_+ one $v_2(r+1)$) and $L > v$, and set $\overline{r} := 2^L - r$ for its T_+ partner. In the conclusions below, write $V_K^{(j)} := \sum_{i<j} v_K^{(i)}$ and $V_I^{(j)} := \sum_{i<j} v_I^{(i)}$ for the cumulative valuations*

along the two tracks; the proof shows that they coincide across the T_+ and T_- reductions, so a single notation suffices. Conclusions:

(i) $\overline{r}$ *is odd with* $v_2(\overline{r}+1) = v < L$ *(so the* T_+ *reduction at* $\overline{r}$ *in the sense of Section 7 uses the same parameter* v*);*

(ii) the T_- *parallel reduction at* (r, L) *and the* T_+ *parallel reduction at* $(\overline{r}, L)$*, both with parameter* v*, terminate at the same index* J*;*

(iii) for every $0 \le j \le J$*,*

$$a_K^{+,(j)} \equiv -\,a_K^{-,(j)} \pmod{2^{L-V_K^{(j)}}}, \quad a_I^{+,(j)} \equiv -\,a_I^{-,(j)} \pmod{2^{L-v-V_I^{(j)}}}, \tag{6}$$

and $v_K^{+,(j)} = v_K^{-,(j)}$*,* $v_I^{+,(j)} = v_I^{-,(j)}$ *whenever* $j < J$*;*

(iv) the affine data satisfy

$$(c_j^+, d_j^+) = (c_j^-, \,-d_j^-) \qquad (0 \le j \le J). \tag{7}$$

Proof. Throughout, a superscript $+$ or $-$ marks the T_+ or T_- reduction and the subscript K or I the two tracks; v_K, v_I are per-step valuations and V_K, V_I their running sums. The I-track modulus carries the extra $-v$ (its residual valuation is $L - v - V_I^{(j)}$, against $L - V_K^{(j)}$ on the K-track) because the I-track starts at level $L - v$.

(i): $\overline{r} + 1 = 2^L - (r-1)$ has, by the ultrametric identity with $v_2(r-1) = v < L$, 2-adic valuation exactly v. Oddness of $\overline{r}$ is immediate.

(iii) and (ii) jointly: we prove the residue congruences in (6) for $j = 0, 1, \ldots$ by induction, and deduce simultaneous termination and the valuation equalities along the way.

Base case ($j = 0$). On the two tracks,

$$a_K^{+,(0)} = \overline{r} = 2^L - r = -a_K^{-,(0)} + 2^L, \qquad a_I^{+,(0)} = (\overline{r}+1)/2^v = 2^{L-v} - a_I^{-,(0)},$$

using $a_I^{-,(0)} = (r-1)/2^v$ and $\overline{r} + 1 = 2^L - (r-1)$. Both congruences in (6) hold at $j = 0$. For the valuation equality at $j = 0$, suppose the T_- reduction is still running, i.e. $v_K^{-,(0)} < L$. Then $3\overline{r} + 1 = 3 \cdot 2^L - (3r - 1)$, whose summands have valuations L and $v_K^{-,(0)}$. Since $v_K^{-,(0)} < L$, the ultrametric identity gives $v_K^{+,(0)} = v_K^{-,(0)}$. The analogous calculation on the I-track gives $v_I^{+,(0)} = v_I^{-,(0)}$.

Inductive step. Assume (6) at index j and that both reductions are still running at index j. By the inductive valuation equalities $v_K^{+,(j)} = v_K^{-,(j)}$ and $v_I^{+,(j)} = v_I^{-,(j)}$ we may write $v_K^{(j)}, v_I^{(j)}$ unambiguously, and the running condition reads $v_K^{(j)} < L - V_K^{(j)}$ and $v_I^{(j)} < L - v - V_I^{(j)}$. For the K-track,

$$3a_K^{+,(j)} + 1 \equiv 3(-a_K^{-,(j)}) + 1 = -(3a_K^{-,(j)} - 1) \pmod{2^{L-V_K^{(j)}}}.$$

The right-hand side has valuation $v_K^{(j)} < L - V_K^{(j)}$, so the left-hand side has the same valuation $v_K^{+,(j)} = v_K^{(j)}$, and dividing through by $2^{v_K^{(j)}}$ gives

$$a_K^{+,(j+1)} = \frac{3a_K^{+,(j)} + 1}{2^{v_K^{(j)}}} \equiv -\frac{3a_K^{-,(j)} - 1}{2^{v_K^{(j)}}} = -\,a_K^{-,(j+1)} \pmod{2^{L - V_K^{(j+1)}}}.$$

The same calculation on the I-track — replacing $L - V_K^{(j)}$ by $L - v - V_I^{(j)}$ and the K-track numerators $3a_K^{\pm,(j)} \pm 1$ by the I-track numerators $3a_I^{\pm,(j)} \pm 1$ — gives the analogous congruence and $v_I^{+,(j)} = v_I^{(j)}$.

Simultaneous termination. The termination condition of the $(\overline{r}, L)$ reduction at index j is $v_K^{+,(j)} \geq L - V_K^{(j)}$ or $v_I^{+,(j)} \geq L - v - V_I^{(j)}$. We show termination at exactly J in two directions. *No earlier stop:* if the T_+ reduction stopped at some $j' < J$, then by the inductive step at j' (valid because both reductions still run at j') the valuation equalities $v_K^{+,(j')} = v_K^{-,(j')}$ and $v_I^{+,(j')} = v_I^{-,(j')}$ hold, so the T_- reduction would also satisfy its stop condition at j', contradicting the definition of J. *Stop at J:* applying the inductive step at $j = J - 1$ — legitimate, since that step needs both reductions to run only at $J - 1$, not at J — yields the residue congruence (6) at $j = J$. Combining this congruence with the T_- K-track stop $3a_K^{-,(J)} - 1 \equiv 0 \pmod{2^{L - V_K^{(J)}}}$ yields

$$3a_K^{+,(J)} + 1 \equiv -(3a_K^{-,(J)} - 1) \equiv 0 \pmod{2^{L - V_K^{(J)}}},$$

so $v_K^{+,(J)} \geq L - V_K^{(J)}$ and the T_+ K-track stops at J; the analogous calculation on the I-track gives $v_I^{+,(J)} \geq L - v - V_I^{(J)}$.

(iv): At $j = 0$, $c_0^+ = 2^{-v} = c_0^-$ and $d_0^+ = +2^{-v} = -\,d_0^-$. Inductively, given $(c_j^+, d_j^+) = (c_j^-, -d_j^-)$ and the valuation equalities $v_K^{+,(j)} = v_K^{-,(j)}$, $v_I^{+,(j)} = v_I^{-,(j)}$,

$$c_{j+1}^+ = c_j^+ \cdot 2^{v_K^{(j)} - v_I^{(j)}} = c_{j+1}^-,$$

$$d_{j+1}^+ = \frac{3d_j^+ - c_j^+ + 1}{2^{v_I^{(j)}}} = -\frac{3d_j^- + c_j^- - 1}{2^{v_I^{(j)}}} = -\,d_{j+1}^-. \qquad \square$$

7.3. **The bijection theorem.**

Theorem 7.3. *For every level L, the involution $r \mapsto \overline{r} := 2^L - r$ restricts to a bijection $\mathcal{O}_L \leftrightarrow \mathcal{O}_L^+$. Consequently $|\mathcal{O}_L^+| = |\mathcal{O}_L|$, and Theorems 1.1 to 1.3 hold verbatim for $\mathcal{O}^+$, with the same density constant c_W and factor complexity function $p_W^+(\ell) = 2^\ell$.*

Proof. For r odd with $v_2(r-1) \geq 1$ and $L > v_2(r-1)$, equation (7) of Lemma 7.2(iv) at the terminal index J together with the sign convention $X_{\text{end}}^+ := c - 3d$ of Remark 7.1 gives

$$X_{\text{end}}^+(\overline{r}, L) = c_J^+ - 3d_J^+ = c_J^- - 3 \cdot (-d_J^-) = c_J^- + 3d_J^- = X_{\text{end}}(r, L).$$

Hence $r \in \mathcal{O}_L$ iff $\overline{r} \in \mathcal{O}_L^+$. Since $r \mapsto \overline{r}$ is an involution on the odd residues in $\{1, \ldots, 2^L - 1\}$, it restricts to a bijection $\mathcal{O}_L \leftrightarrow \mathcal{O}_L^+$. Concretely, at $L = 5$ this gives $\mathcal{O}_5^+ = \{5\}$, and at $L = 6$ it gives $\mathcal{O}_6^+ = \{2^6 - r : r \in \mathcal{O}_6\} = \{5, 37, 45\}$, the anchor set parallel to $\mathcal{O}_6$ in the T_+ analogue of the factor-complexity construction below.

For the remaining claims, observe that Lemma 7.2 conjugates each T_- basic step into the corresponding T_+ basic step.

Lift. The T_+ lift theorem (the analogue of Theorem 1.1) follows by composing the involution with the T_- lift. Given $\overline{r_0} \in \mathcal{O}_{L_0}^+$, set $r_0 := 2^{L_0} - \overline{r_0} \in \mathcal{O}_{L_0}$; by Theorem 1.1, both r_0 and $r_0 + 2^{L_0}$ lie in $\mathcal{O}_{L_0+1}$. Their level-$(L_0 + 1)$ involution images

$$2^{L_0+1} - r_0 = \overline{r_0} + 2^{L_0}, \qquad 2^{L_0+1} - (r_0 + 2^{L_0}) = \overline{r_0}$$

lie in $\mathcal{O}_{L_0+1}^+$ by the level-(L_0+1) version of the bijection. Hence both T_+-lifts of $\overline{r_0}$ are again T_+-obstructions. Positive density and full factor complexity for $\mathcal{O}^+$ then propagate exactly as in Sections 4 and 5, with $\mathcal{O}^+$ in place of $\mathcal{O}$ throughout the argument. $\square$

8. Open questions

8.1. **The exact value of c_W.** The principal open question is to determine c_W exactly, or equivalently to give a quantitative bound on the growth of $|\mathcal{A}_L^{G\neq 0}|$. Three approaches present themselves, each requiring substantial additional work.

(i) Substitution-theoretic. Stérin [Sté20] realizes T_+ as a four-state finite-state transducer between bases 2 and 3. An analogous construction for the parallel reduction, combined with Pansiot's classification [Pan84] (refined by Devyatov [Dev15]), would determine the asymptotic growth class of $\mathcal{A}_L^{G\neq 0}$. The relevant question concerns *non-primitive* substitution shifts: the primitive case is ruled out by Corollary 5.2, so any realisation must lie outside that classification (Remark 5.3). The natural setting for such a realisation is the S-adic framework [BD14].

(ii) Markov-operator-spectral. Wirsching [Wir98] constructs a Markov operator W_3 on residue-density functions for the T_+ predecessor problem. An analogous operator for the parallel reduction would have an action on (c, d)-space; its spectrum would determine the asymptotic growth of the shift-class populations $|\mathcal{O}_L^{G_s}|$. A unified treatment — for instance, expressing c_W in terms of Wirsching's invariant density φ — appears not to be in the literature.

(iii) Generating-function-combinatorial. A direct enumeration via generating functions in L is in principle accessible, but the absence of a closed-form recursion — the count depends on the joint distribution of stopping times and shift indices — makes the analysis delicate.

8.2. **Nontrivial cycles.** The shift-zero class $\mathcal{O}_L^{G_0}$ encodes residues for which the two parallel stopping times remain rigidly tied. Whether the existence (or absence) of additional nontrivial T_--cycles is reflected in the structure of $\mathcal{O}_L^{G_0}$ for large L — in particular, whether one can extract a Diophantine obstruction from the strict lift balance of Corollary 6.4 — is a natural and apparently open question. Classical lower bounds on the length of nontrivial T_+-cycles, due to Eliahou [Eli93] and extended by Belaga and Mignotte [BM06], indicate how rigidly the Diophantine geometry constrains cycle structure; an analogous bound in terms of $\mathcal{O}_L^{G_0}$ would be of independent interest.

Extension to the $(3n + b)$ family

The constructive lift, the positive-density bound, and the full-factor-complexity result extend to the family $T_b(n) := (3n + b)/2^{v_2(3n+b)}$ for every odd b. The bridge is a multiplicative residue-bijection on $\{1, \dots, 2^L\}$ that conjugates the obstruction structure across the family and preserves the shift index; the involution $r \mapsto 2^L - r$ used in Section 7 is a special case. A detailed treatment is in preparation as a separate paper.

Further extensions

We have also established significant partial results for general multiplier a — in particular structure theorems for $a = -1$ and for the negative-Mersenne family $a = -(2^v - 1)$ with explicit lower bounds on c_W. A complete treatment of the $(an+b)$ family raises new phenomena (Mersenne asymmetry, a sub-class hierarchy, a signed sync index) and is the subject of ongoing work.

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Use of AI tools

This manuscript was prepared with substantial assistance from generative AI tools. Anthropic's Claude assisted with developing and formulating the proofs, drafting and refining the exposition, and writing the accompanying verification code; OpenAI's ChatGPT was used additionally for independent second-opinion reviews of the arguments. All definitions, theorems, proofs, and results were reviewed and independently checked by the author and are backed by the reproducible verification and falsification code described in Appendix A. The author takes full and sole responsibility for the correctness and content of this manuscript. No AI system is listed as an author.

Appendix A. Computational verification

The algorithmic X-invariant criterion of Section 2 is implemented in Python in the supporting code that accompanies this paper, collected in the directory `code/python/` alongside the manuscript source and mirrored at `https://github.com/yaccob/collatz/tree/main/manuscripts/obstruction_residues/code`. A permanent archival snapshot is deposited on Zenodo under the concept DOI `10.5281/zenodo.20356496`, which resolves to the most recent archived version. The directory contains the verification scripts for each rigorous result (lift theorem, atom decomposition, the no-shift-zero-atom theorem (Theorem 6.2), full factor complexity, simultaneous stop) together with a `README.md` mapping each script to the result it certifies. For each level $L \le L_{\max}$, the relevant script enumerates all odd residues r with $v_2(r-1) \ge 1$, runs the parallel reduction, and records the terminal data. The set $\mathcal{O}_L$ is read off as the preimage of $\{(c,d) : 3d+c=1\}$.

We have run this enumeration through $L=16$ in Python, producing the counts of Table 3. A second, independent Rust implementation (`code/rust/`) reproduces these counts for $L=5,\dots,16$ and extends the rigorous density bounds of Table 1 to $L=32$.

L	$\lvert\mathcal{O}_L\rvert$	$\lvert\mathcal{O}_L^{G_0}\rvert$	$\lvert\mathcal{O}_L^{G\neq 0}\rvert$	$\lvert\mathcal{A}_L^{G\neq 0}\rvert$
5	1	0	1	1
6	3	1	2	1
7	8	4	4	2
8	19	12	7	3
9	42	31	11	4
10	91	73	18	7
11	194	164	30	12
12	409	358	51	21
13	855	767	88	37
14	1776	1622	154	66
15	3671	3398	273	119
16	7556	7069	487	214

Table 3. Direct enumeration counts at levels $L=5,\dots,16$: total obstructions, shift-zero and shift-nonzero subclasses, and atomic shift-nonzero obstructions. The decomposition $|\mathcal{O}_L| = |\mathcal{O}_L^{G_0}| + |\mathcal{O}_L^{G\neq 0}|$ is read off each row, and the strict lift balance $|\mathcal{O}_L| = 2|\mathcal{O}_{L-1}| + |\mathcal{A}_L^{G\neq 0}|$ (Corollary 6.4) holds row-by-row against the previous one for $L \ge 6$.

The data satisfies the strict lift balance $|\mathcal{O}_L| = 2|\mathcal{O}_{L-1}| + |\mathcal{A}_L^{G\neq 0}|$ of Corollary 6.4 at every level, and is consistent with Theorems 1.1, 6.2 and 6.5 (series convergence).

The rigorous bound $c_W \geq 7556/65536 \approx 0.115$ of Theorem 1.2 is conditional on the correctness of this enumeration. The check is self-validating in a useful sense: the strict lift balance of Corollary 6.4 is asserted row-by-row by the script `count_obstructions.py` against the previous-level count, so any drift in the enumeration would manifest as an immediate hard failure rather than silently shifting the bound. Each rigorous-verification script exits non-zero on any violation of the asserted theorem (a convention checked statically by `meta_test_exit_convention.py`), so the bound is reproducible end-to-end by running the script.

A.1. **The shift-index window.** The depth bound (Theorem 6.13) and the resulting window (Corollary 6.14) are certified by the same convention: each script below asserts its claim and exits non-zero on violation. The two analytic regimes reduce to elementary arithmetic on the closed forms — `depth_bound_threshold.py` confirms $f(2a+2) < 1 \iff a \geq 4$ for (N) (Lemma 6.9), and `depth_closed_form.py` verifies the extremal sums $A_K^{\min}, A_I^{\max}$ against a brute optimisation over the structural families, the factorisation $D = C_0 + 2^{v+q}\Gamma$, and the positivity ($\Gamma > 0$, $\Phi > 3$, $D(J, s, v, 0) > 0$) on the window $1 \leq v \leq J \leq 2s + 2$ for $s \geq 4$ (Lemma 6.10). The sign-free identity (4) (Lemma 6.8) is checked for every obstruction up to $L = 19$ by `signfree_identity_check.py` (0 violations over 126648 obstructions, of which 1881 have $s < 0$, so both signs are exercised), and `shift_index_window_check.py` confirms Theorem 6.13 and Corollary 6.14 directly over all obstructions up to $L = 20$.

The small-shift tails $|s| \in \{2, 3\}$ (Lemma 6.12) are the finite computations. For each carry-automaton stratum (Lemma 6.11) the reachable-set sequence $R_m = T_A^m(R_0)$ reaches a genuine fixed point (period $P = 1$) after a short preperiod M_0, on a small reachable state space; since the period is 1, a single fixed point decides the stratum for every $q \geq 0$. Table 4 lists the certificate, maximised over $1 \leq v \leq J$.

The residual cells left by the closed form on the positive side are closed by exhaustive search over the coupled valuation sequences (`depth_dfs_tails.py`): the stratum $s = 3$ leaves ($J = 8$, $q = 0$, $v \in \{2, \ldots, 8\}$) — 4096 length-8 sequences, 0 obstructions — and $s = 2$ leaves ($J = 5$, $q \in \{0, 1\}$) — 1443 length-5 sequences, 0 obstructions. Both searches find genuine obstructions at the tight depth $J = 2s + 3$ (e.g. $r = 159749 \in \mathcal{O}_{18}^{G_3}$ at $J = 9$), confirming the enumerations are not vacuous.

stratum (s,J)	preperiod M_0	period P	reachable states
$(2,6)$	≤ 25	1	≤ 185
$(-2,5)$	≤ 16	1	≤ 88
$(-2,6)$	≤ 25	1	≤ 185
$(-3,8)$	≤ 35	1	≤ 777

Table 4. Fixed-point certificates for the carry-automaton strata of Lemma 6.11, maximised over $1 \le v \le J$. The positive stratum $(2,6)$ is certified by `depth_carry_automaton_k2.py`, the negative strata by `depth_carry_automaton_neg.py`. Every period is $P = 1$, so a single fixed point rejects every $q \ge 0$, and the reachable state space is small enough (≤ 777 nodes) to enumerate explicitly. The reachable-state count for $(2,6)$ is verified independently by `depth_carry_statespace.py`, which also exhibits the invariant carry interval $[-81, 81]$ entered within two block-A steps. The stratum $(-2,6)$ reproduces the state table of $(2,6)$, a consequence of the sign-free form of (4).

References

[AS03] Jean-Paul Allouche and Jeffrey Shallit, *Automatic sequences: Theory, applications, generalizations*, Cambridge University Press, 2003.

[BD14] Valérie Berthé and Vincent Delecroix, *Beyond substitutive dynamical systems: S-adic expansions*, RIMS Kôkyûroku Bessatsu **B46** (2014), 81–123.

[BM06] Edward G. Belaga and Maurice Mignotte, *Walking cautiously into the Collatz wilderness: Algorithmically, number-theoretically, randomly*, Fourth Colloquium on Mathematics and Computer Science: Algorithms, Trees, Combinatorics and Probabilities, Discrete Mathematics and Theoretical Computer Science Proceedings AG, 2006, pp. 249–260.

[BR10] Valérie Berthé and Michel Rigo (eds.), *Combinatorics, automata and number theory*, Encyclopedia of Mathematics and its Applications, vol. 135, Cambridge University Press, 2010.

[Cas97] Julien Cassaigne, *Complexity and special factors*, Bulletin of the Belgian Mathematical Society — Simon Stevin **4** (1997), no. 1, 67–88.

[Cra78] R. E. Crandall, *On the "$3x+1''$ problem*, Mathematics of Computation **32** (1978), no. 144, 1281–1292.

[Dev15] Rostislav Devyatov, *On the subword complexity of morphic words*, Acta Informatica **52** (2015), no. 4, 379–425.

[DHS99] Fabien Durand, Bernard Host, and Christian Skau, *Substitutional dynamical systems, Bratteli diagrams and dimension groups*, Ergodic Theory and Dynamical Systems **19** (1999), no. 4, 953–993.

[Eli93] Shalom Eliahou, *The $3x+1$ problem: new lower bounds on nontrivial cycle lengths*, Discrete Mathematics **118** (1993), 45–56.

[Eve77] C. J. Everett, *Iteration of the number-theoretic function $f(2n) = n$, $f(2n+1) = 3n+2$*, Advances in Mathematics **25** (1977), no. 1, 42–45.

[Fog02] N. Pytheas Fogg, *Substitutions in dynamics, arithmetics and combinatorics*, Lecture Notes in Mathematics, vol. 1794, Springer, 2002.

[Kor94] Ivan Korec, *A density estimate for the $3x + 1$ problem*, Mathematica Slovaca **44** (1994), no. 1, 85–89. MR 1290275

[Lag85] Jeffrey C. Lagarias, *The $3x + 1$ problem and its generalizations*, American Mathematical Monthly **92** (1985), 3–23.

[Lag10a] ———, *The $3x + 1$ problem: an annotated bibliography (1963–1999) and (2000–2009)*, 2010, arXiv:math/0309224 and arXiv:math/0608208.

[Lag10b] Jeffrey C. Lagarias (ed.), *The Ultimate Challenge: The $3x+1$ Problem*, American Mathematical Society, 2010.

[LM95] Douglas Lind and Brian Marcus, *An introduction to symbolic dynamics and coding*, Cambridge University Press, 1995.

[Lot02] M. Lothaire, *Algebraic combinatorics on words*, Encyclopedia of Mathematics and its Applications, vol. 90, Cambridge University Press, 2002.

[MH38] Marston Morse and Gustav A. Hedlund, *Symbolic dynamics*, American Journal of Mathematics **60** (1938), no. 4, 815–866.

[MH40] ———, *Symbolic dynamics II. Sturmian trajectories*, American Journal of Mathematics **62** (1940), no. 1, 1–42.

[Pan84] Jean-Jacques Pansiot, *Complexité des facteurs des mots infinis engendrés par morphismes itérés*, Automata, Languages and Programming (ICALP 1984), Lecture Notes in Computer Science, vol. 172, Springer, 1984, pp. 380–389.

[Sté20] Tristan Stérin, *The Collatz process embeds a base conversion algorithm*, 2020.

[Tao22] Terence Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum of Mathematics, Pi **10** (2022), Paper No. e12.

[Ter76] Riho Terras, *A stopping time problem on the positive integers*, Acta Arithmetica **30** (1976), no. 3, 241–252.

[Wir98] Günther J. Wirsching, *The dynamical system generated by the $3n + 1$ function*, Lecture Notes in Mathematics, vol. 1681, Springer, 1998.

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